

Watsonx Orchestrate

Level 4 lab prep guide – Sept 2025 version



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Introduction

IBM lab environments are accessed directly from the course. Each course that includes a lab component has an activity within the course that allows you to easily access your own lab environments.

Each lab environment is specific to the product or use cases that are used in the course material. However, the basic steps for reserving and provisioning an environment are described in this document.

It can take some time for the lab environment to be ready for use. In some cases, it can be up to 90 minutes. In other cases, it might be less. After reserving your environment on TechZone, watch your mailbox. You will receive several emails about the status of your reservation (Provisioning / ready) or emails to activate IBM Cloud account.

The environments will be available to you for **5 days** from the time that you reserve it. Time can be extended one time following the IBM TechZone reservation policy.

Create an account to request your environment

IBM employees can use their IBM credentials to access IBM Technology Zone or IBM Cloud.

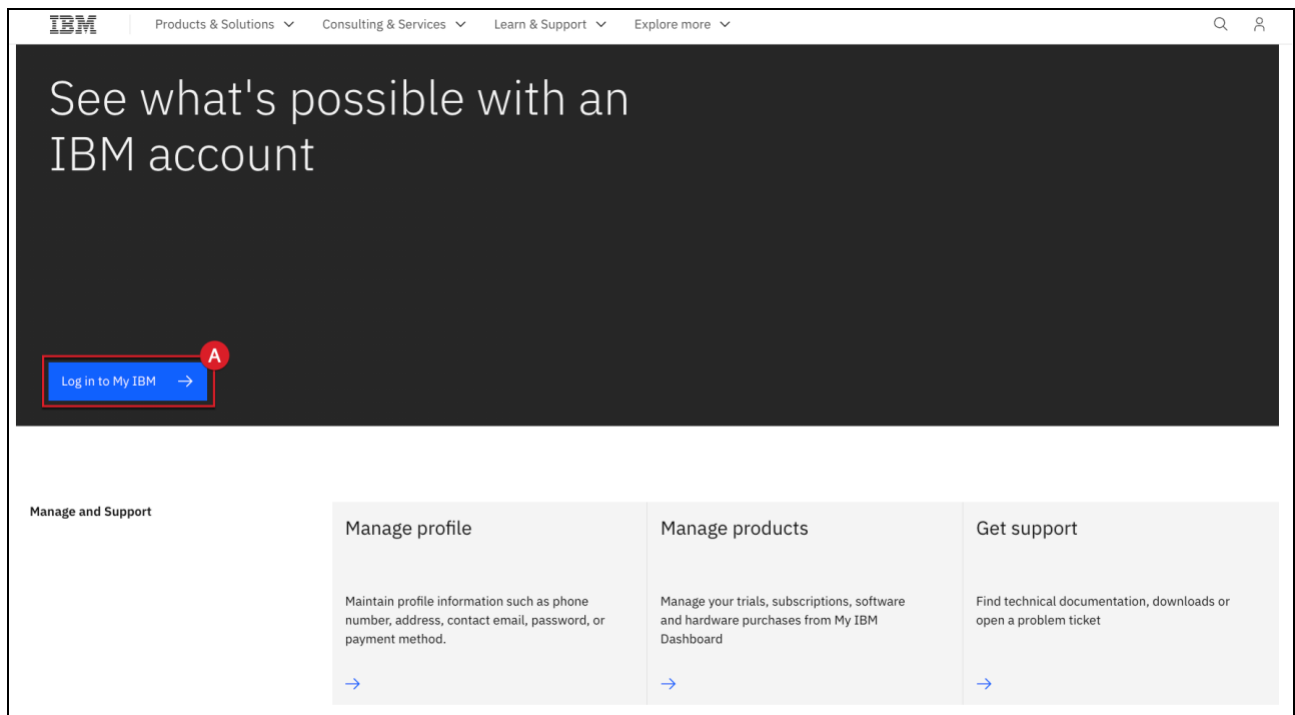
Business Partners must use an IBMid for IBM Technology Zone. If you do not have an IBMid, you can create one by using steps in this document. There is no cost to creating an IBMid.

You might also require an IBM Cloud account for some lab environments.

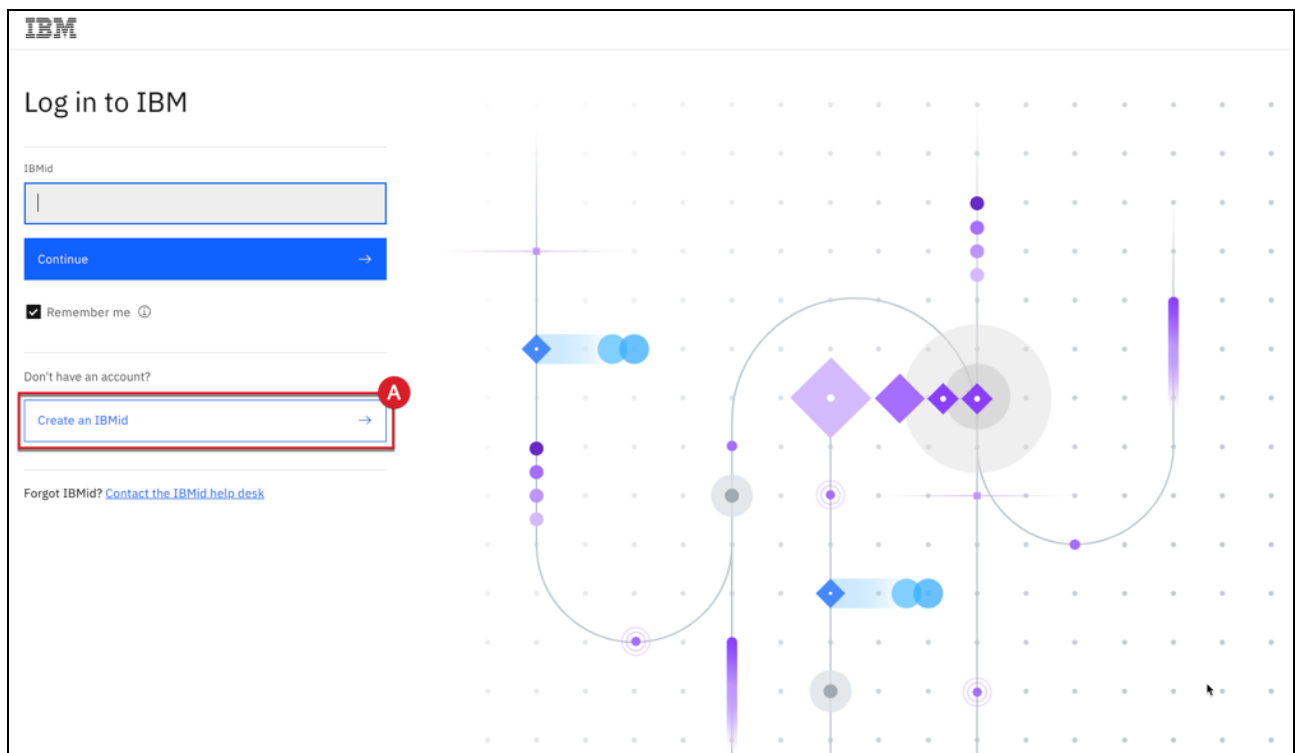
Additionally, you might need to configure an OpenVPN connection to be able to access the lab environment.

Creating an IBMid

1. In your web browser, go to the IBM account page (<https://www.ibm.com/account/us-en/>).
1. Click **Log in to My IBM** (A).



2. The Log in to IBM page is displayed. Click **Create an IBMid** (A).



3. On the **Create an IBMid** page, enter the following information:

- Your email address (A)
- A password for your IBMid account (B)
- First name (C)
- Last name (D)
- Your country or region of residence (E) and state or province (F)
- If you select that you are not a student (G), then enter the name of your company (G)
- Click **Next** (I) to continue

The screenshot shows the 'Create an IBMid' page. On the left, a 'Welcome to IBM' message says 'Create an account to access trials, demos and services.' The main form is titled 'Create an IBMid' and includes a link for users who 'Already have an IBM account? Log in'. The form is divided into sections: 'Account information' and 'Personal information'. The 'Account information' section contains 'Business email' (labeled A) and 'Password' (labeled B). The 'Personal information' section contains 'First name' (labeled C), 'Last name' (labeled D), 'Country or region of residence' (labeled E), and 'State or province' (labeled F). Below these is a section for 'Are you a student?' with 'Yes' and 'No' radio buttons (labeled G). The 'No' option is selected. Below this is a 'Company' section with a text box (labeled H) containing 'IBM' and a 'Next' button (labeled I).

Welcome to IBM

Create an account to access trials, demos and services.

Create an IBMid

Already have an IBM account? [Log in](#)

Account information

Business email ☒

We'll send you a 7 digit code that you'll use to verify your email in step 2.

Password ☒

First name

Last name

Country or region of residence

State or province

Are you a student?

☐ Yes ☒ No

Company

4. You are then prompted to verify your email and complete your account creation. Enter the code that was emailed to you in **Verification token** (A), and then click **Create account** (B).

Welcome to IBM

Create an account to access trials, demos and services.

Create an IBMid

Already have an IBM account? [Log in](#)

Account information

Verify email

We emailed a 7 digit code to [itzuser@email.com](#). This code will expire in 30 minutes.

Verification token

554866d

Didn't receive the email? Check your spam filter for an email from [ibmacct@iam.ibm.com](#).

You can resend the verification token in 43 seconds

I'd like IBM to use my contact details to keep me informed about products, services, and offers. More information on how IBM uses data and ways to [opt out](#) can be found in the [IBM Privacy Statement](#).

☒ Email

I accept the product [Terms and Conditions](#).

Create account

5. The **About your IBMid Account** page is shown. Click **Proceed** (A).

IBM

Welcome to IBM

Create an account to access demos and services.

About your IBMid Account

By creating, or using, an IBMid to access an IBM Web site or other service, you acknowledge that you have read, understand, and agree to be bound by the below terms, in addition to the terms in the IBM Terms of Use, the IBM Privacy Agreement and the "About your IBMid Account Privacy" section of this notice provide details on how IBM protects your personal data.

Most IBMid accounts are individual IDs, meaning you manage your own account information and access your account through the IBMid login service (i.e. IBMid is the authenticating agency).

If your account is created using an email address containing a domain owned by an organization which you are employed by, contracted to, or volunteer for, your organization can:

- inquire about the status of your ID
- request your account settings (including your personal information)
- at its option, convert it to an enterprise ID.

If your account was created as (or is converted to) an enterprise ID, your organization manages your account information. It is also the authenticating agency for your ID, meaning your access is controlled by your organization's login service.

About your IBMid Account Privacy

What data does IBM collect?

Why IBM needs your data

How your data is obtained

How IBM uses your data

How IBM protects your data

How long we keep your data

Your rights

I acknowledge that I understand how IBM is using my Basic Personal Data and (if applicable) how my organization may become the authenticating agency for my IBMid account. I certify that my age is at least 18 years old.

Cancel

Proceed

- After a few minutes, you will receive an email confirming that your IBMid has been activated.

Note: The first time that you log in using your IBMid you will be sent another code to verify your log in.

Creating an IBM Cloud Account (optional)

If you already have an IBM Cloud account, you can skip this task.

- Go to IBM Cloud (<https://cloud.ibm.com/registration>).
- Enter your email address (A) and a password (B), and then click **Next** (C).

The screenshot shows the 'Create an IBM Cloud account' page. At the top, there are links for 'Catalog', 'Cost estimator', and 'Docs'. The main heading is 'Create an IBM Cloud account', with a link 'Already have an IBM Cloud account? Log in' below it. The registration form is divided into sections: 'Account information', 'Verify email', 'Personal information', and 'Account notice'. The 'Account information' section contains an 'Email' field (labeled A), a 'Password' field (labeled B), and a 'Next' button (labeled C). There is also a 'Sign up with Google' button. To the right of the form, there are three benefits listed: 'Unlock the full IBM Cloud Catalog', '200 USD Cloud Credit for free', and 'Get started for free'. At the bottom of the form, there is a 'Create account' button.

After you complete the registration, you receive an email to activate your account. This can take a few hours to process. After you receive the confirmation email, follow the instructions provided in the email to activate your account.

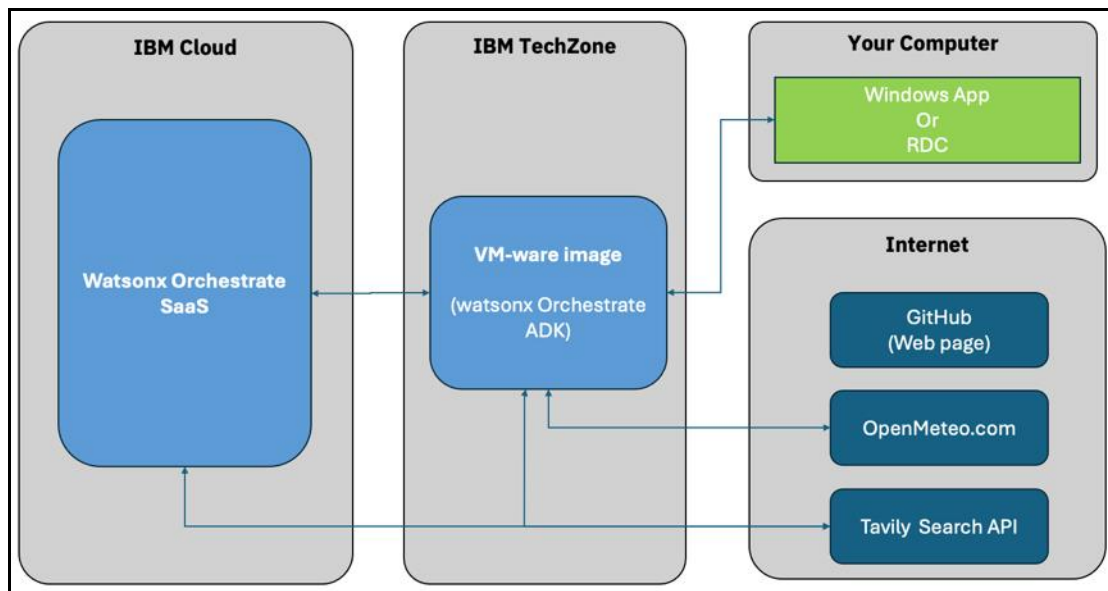
About this lab

This preparation guide provides you with all the step-by-step instructions to get and configure the different environments or accounts that are required to run all the watsonx Orchestrate L4 lab activities.

High level architecture

This diagram gives you a high-level overview of the required environments to perform the watsonx Orchestrate Level 4 learning plan. You will need to reserve two environments on IBM TechZone:

- A watsonx Orchestrate SaaS hosted on IBM Cloud
- A pre-configured VM-ware image containing watsonx Orchestrate ADK



This guide will drive you through the different steps to provision these environments or create the 3rd party product accounts (Tavily in particular).

Lab credentials

During the watsonx Orchestrate L4 learning plan, you will navigate between different products, environments, accounts and services. This prep guide will help you to create this note to collect and reuse the different user/passwords, tokens and API keys you will need during the various lab activities.

You can copy this table in a separate note and use it to store the different element as you go through the preparation instructions.

Environment	Credential	Value	Comments	Item #
ADK VMWare image	User	Administrator	Note: For MS users you may have to use \\Administrator as the username.	Item 1
	Password	IBMDem0s		Item 2
	RDC URL		Remote Desktop URL	Item 3
Watsonx Orchestrate SaaS	Account name		The ibm cloud account name you have been onboarded onto.	Item 6
	Service URL		The service url of your watsonx Orchestrate SaaS instance	Item 7
	API KEY		The API key you have generated to access your watsonx Orchestrate SaaS tenant	Item 8
Tavily Web Search (Used to create an agentic tool)	Command line	<code>npx -y tavily-mcp@0.1.3</code>		Item 10
	TAVILIY_API_KEY			Item 11

Required Tools and Accounts

Remote Desktop (Mac users only)

Access to the virtual machine will be through a remote desktop connection, so you will also need a suitable RDP client installed on your computer.

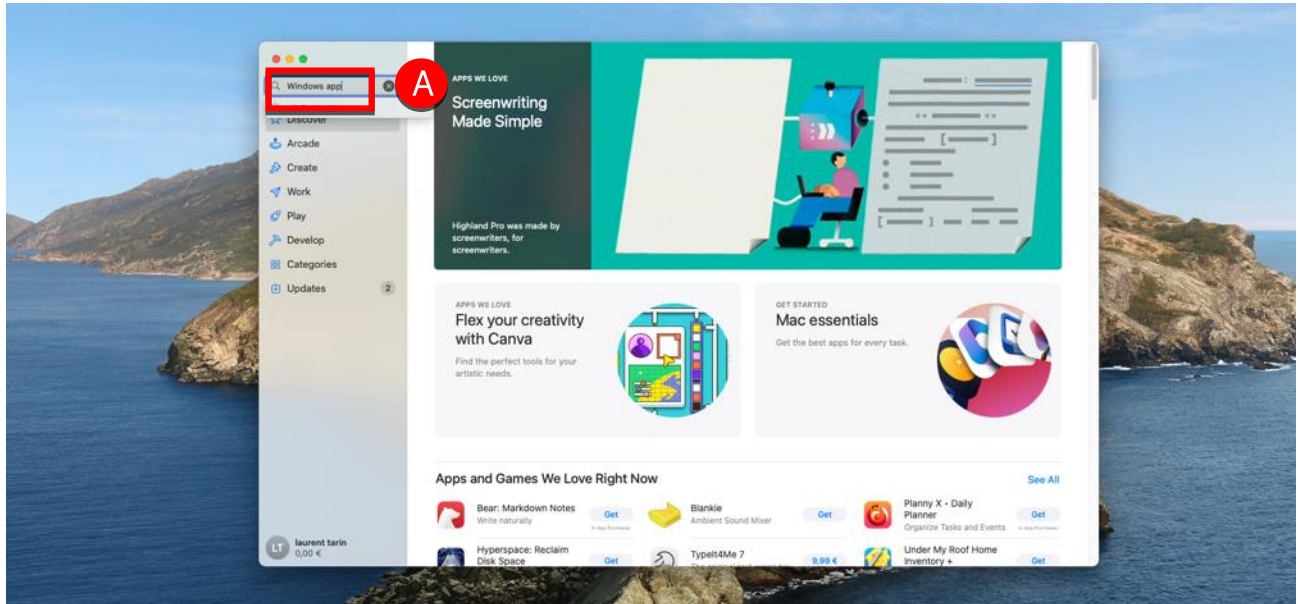
For Windows users, this is already included in the operating system, but for Mac users we recommend you download the Windows App (Microsoft Remote Desktop application) from the AppStore, if you do not already have it installed.

For Mac users, please proceed with the following steps to install Windows App. Windows users can move to the following section.

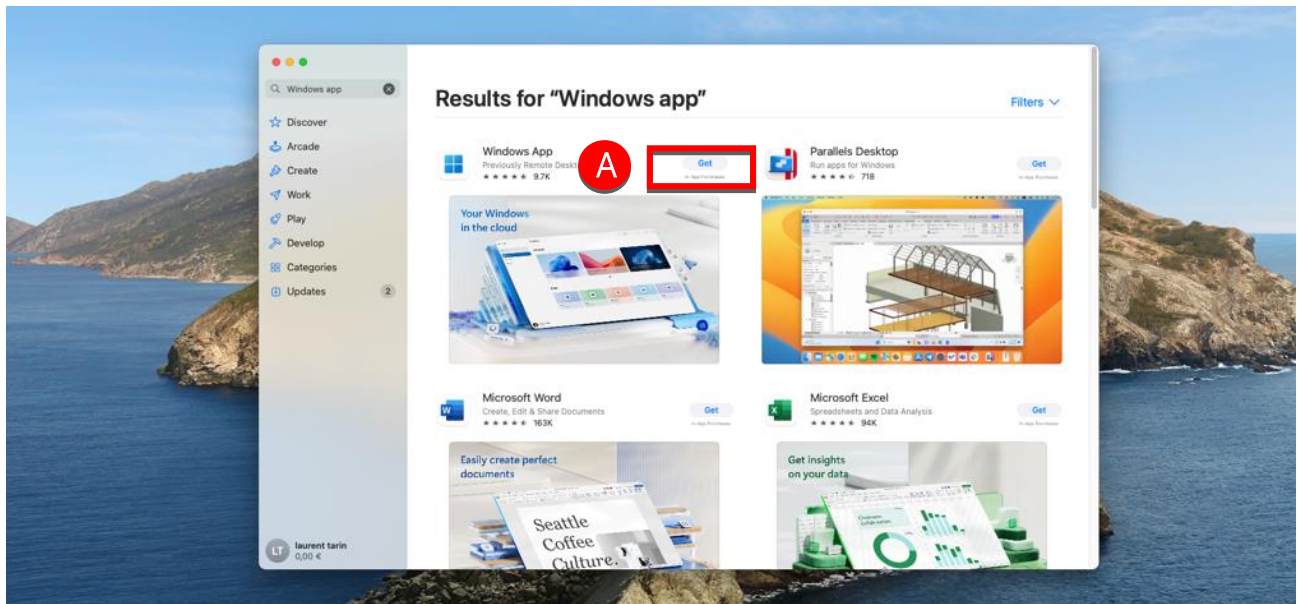
3. Click the **App Store** app icon (A).



4. Type “Windows App” in the search menu (A).



5. Click **Get** next to the Windows App tile (A).



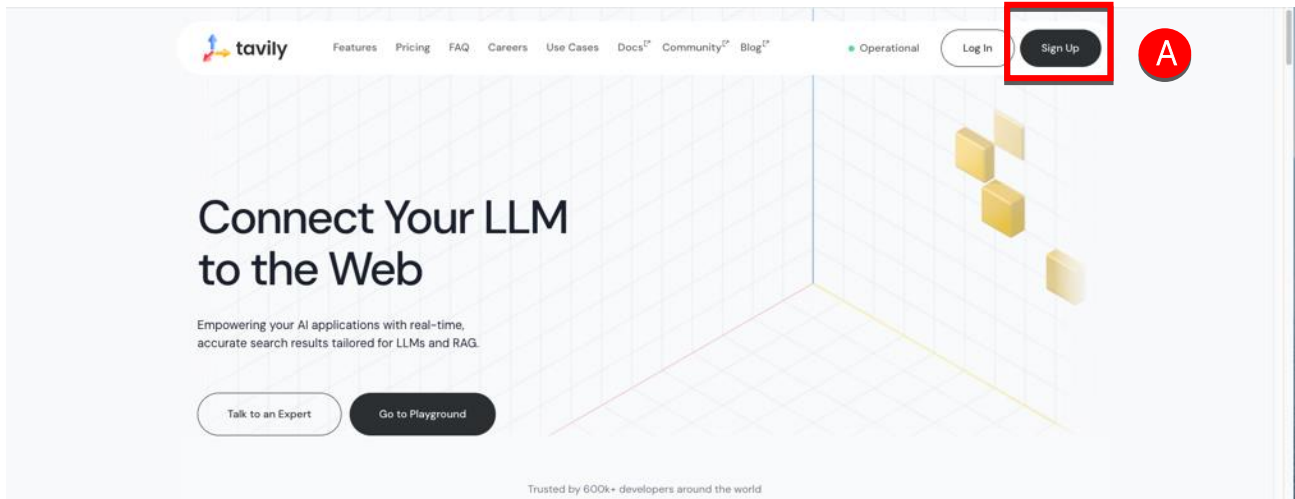
Follow the Windows App installation instructions until the app is installed on your computer.

Tavily

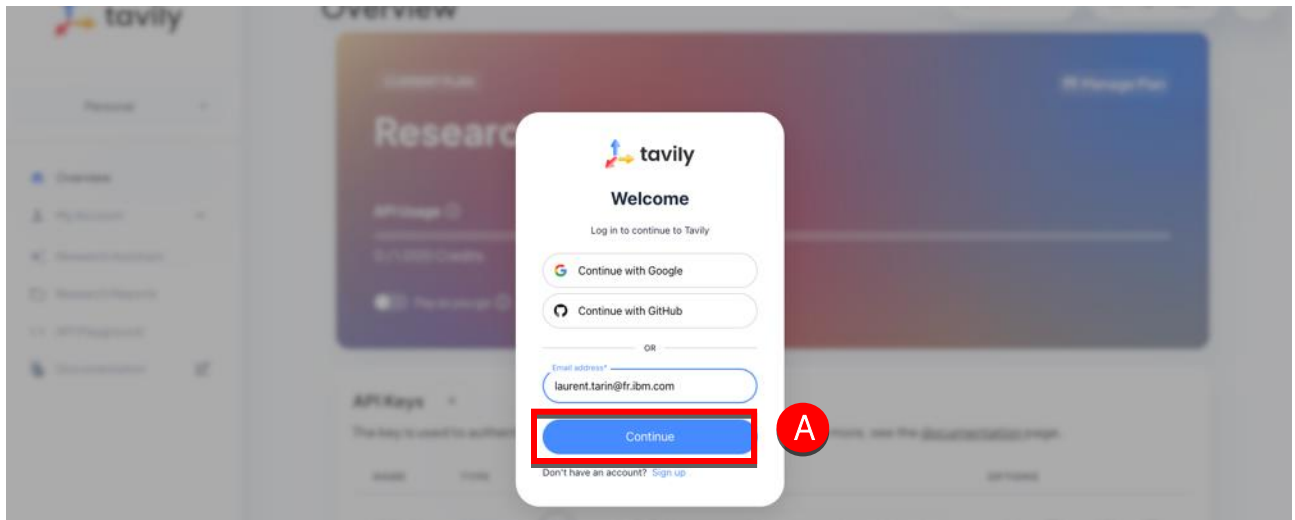
Create a free account

Tavily MCP server requires API keys. To generate this API key, you must first create a free-of-charge account on tavily.com.

1. In your web browser, go to <https://www.tavily.com/>.
2. Click **Sign-Up** (A).

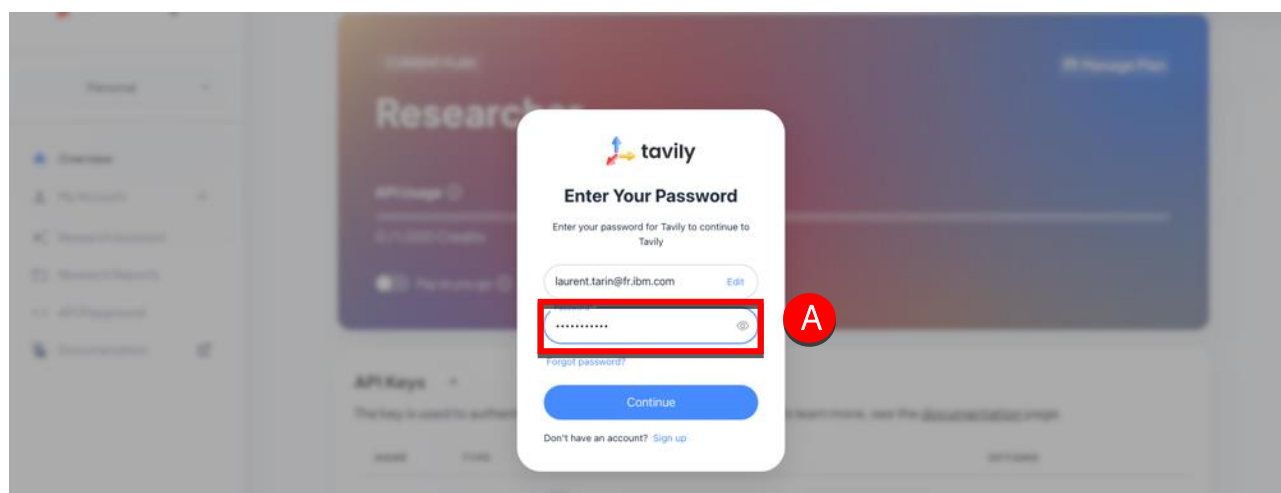


3. Enter your email address (A).

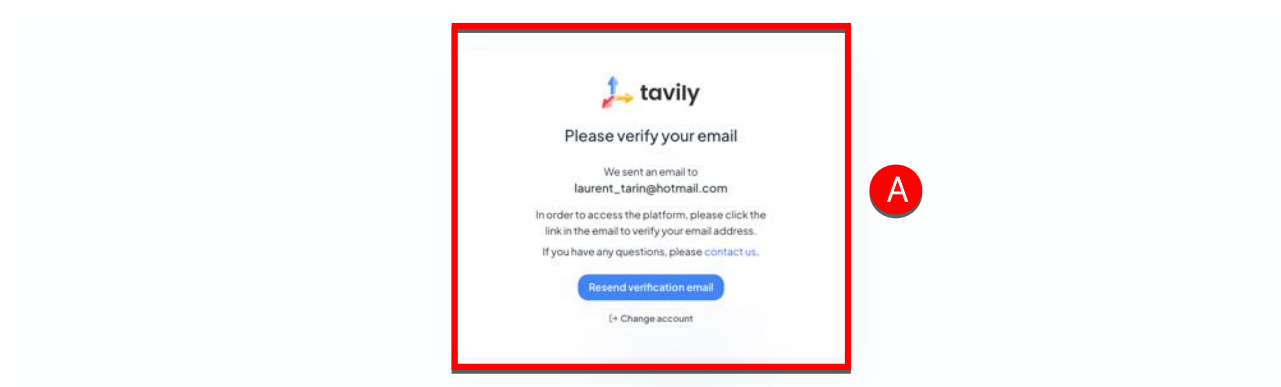


Note: If you have a google account, click Continue with google and follow the instructions to grant access from your google account.

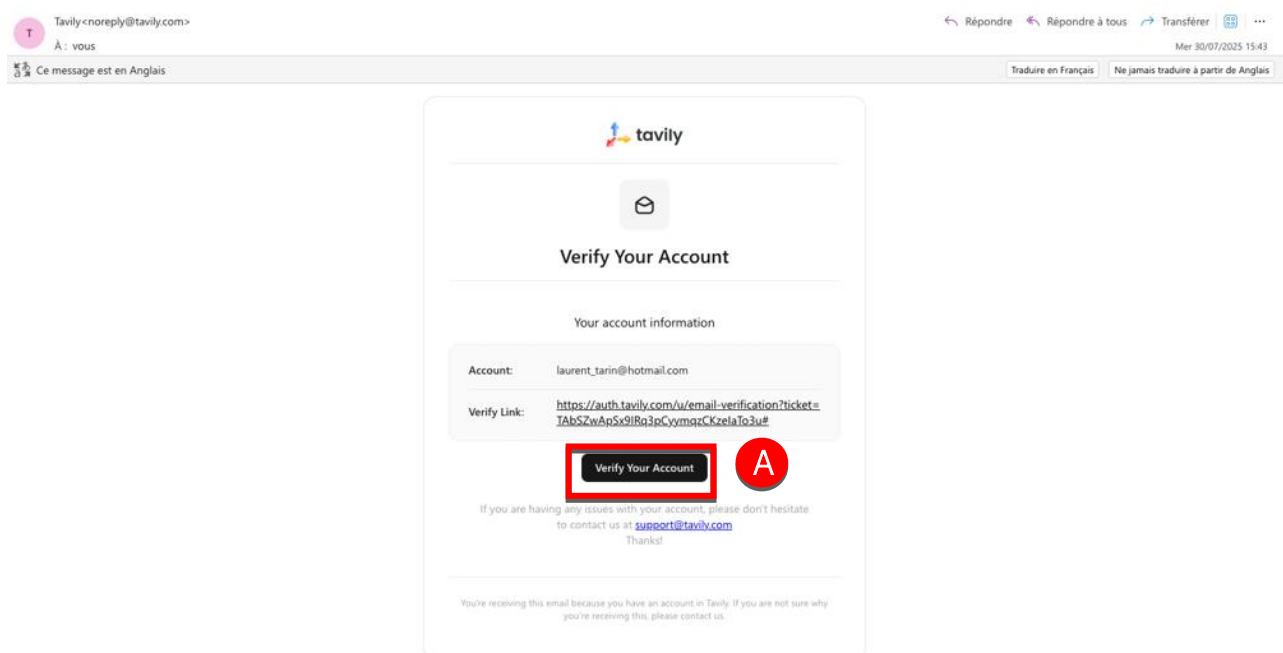
4. Enter a password (A).



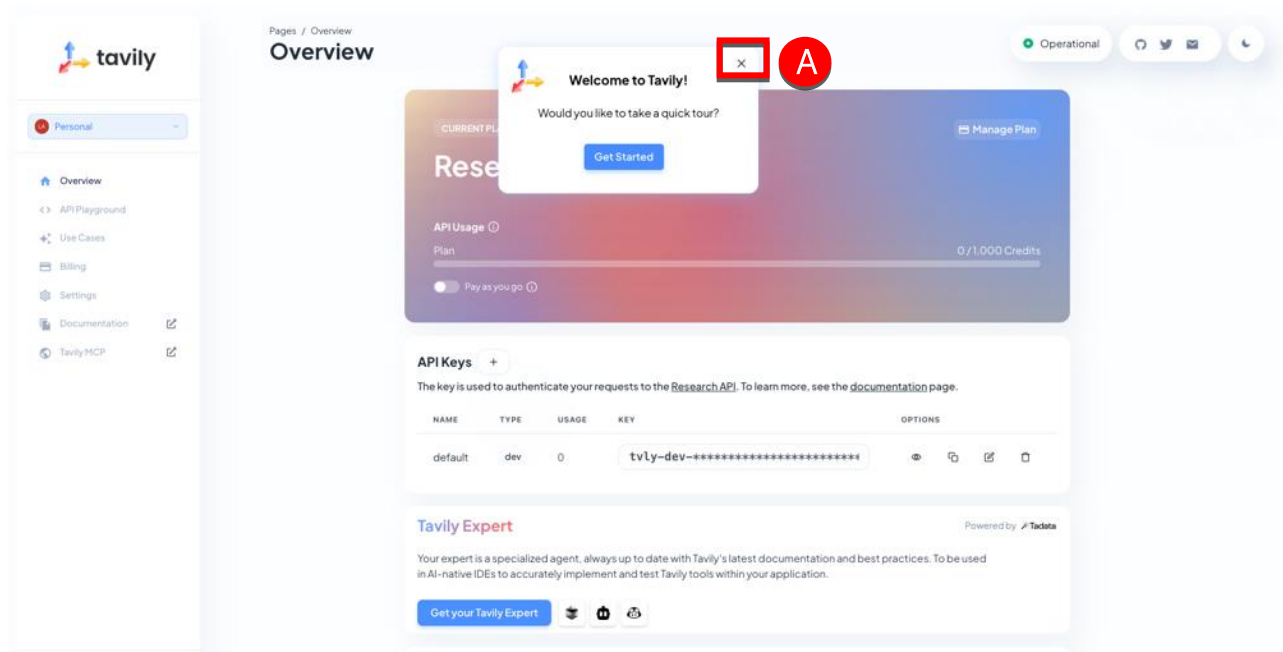
5. On this message (A), go to your corresponding email box to verify your email.



6. Open the **Verify Your Account** email received from Tavily and click **Verify Your account** (A).

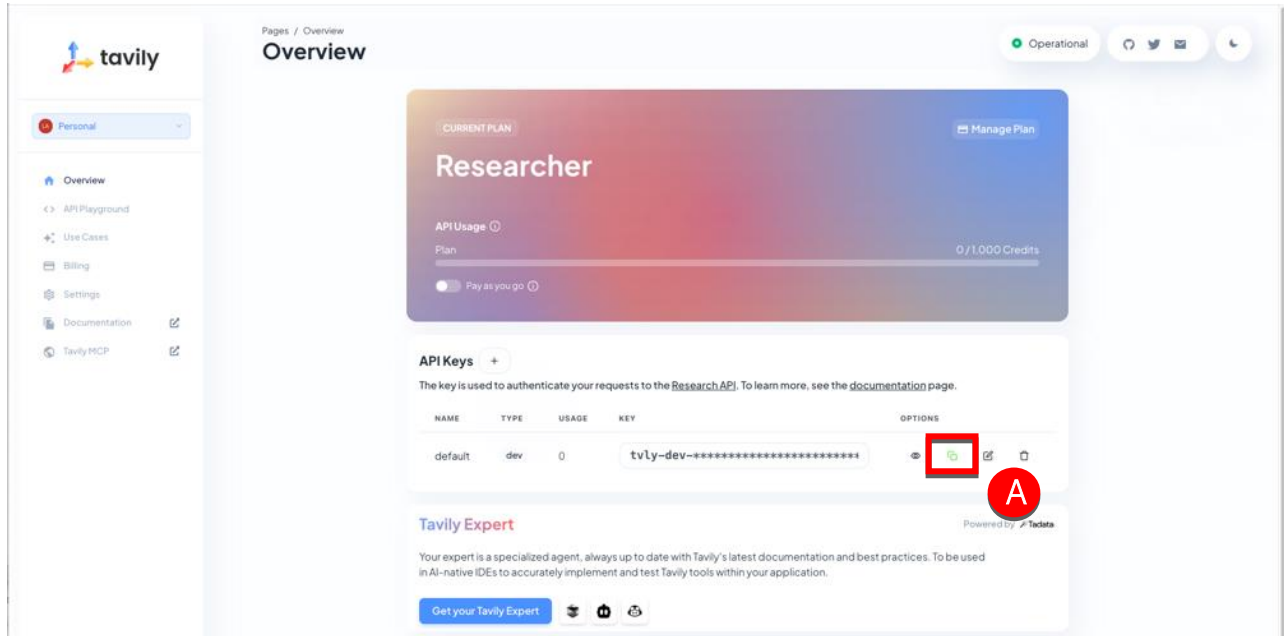


7. From the Tavily page close the **Welcome to Tavily** window (A).



Get your API key

8. A default API Key has been created for you. Click the copy to clipboard icon (A) and save the Tavily API key in your notes.

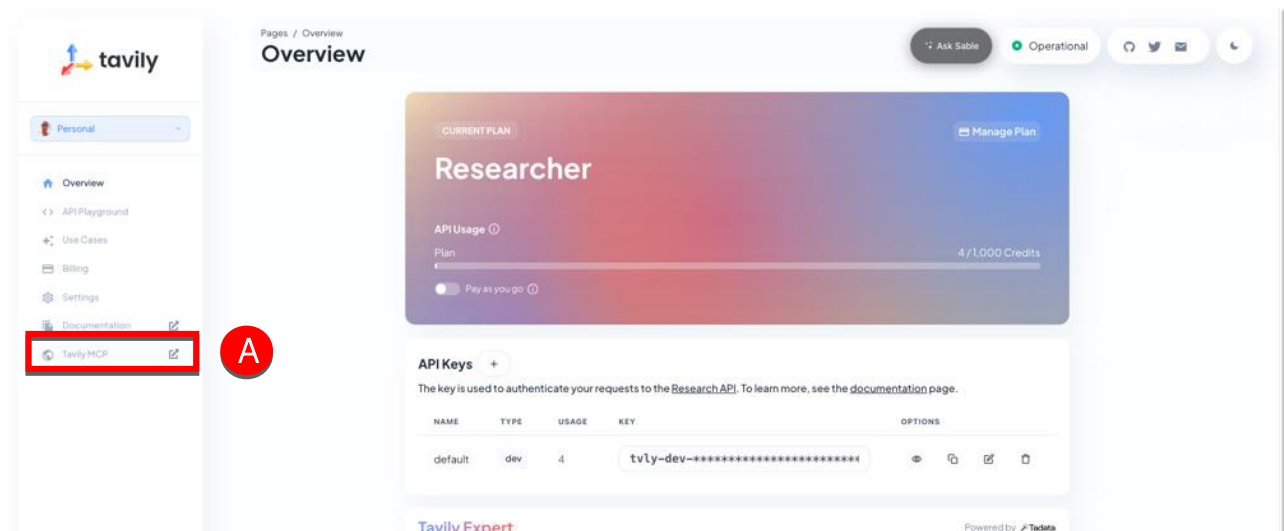


This API key you have saved will be used to configure the MCP Server in watsonx Orchestrate.

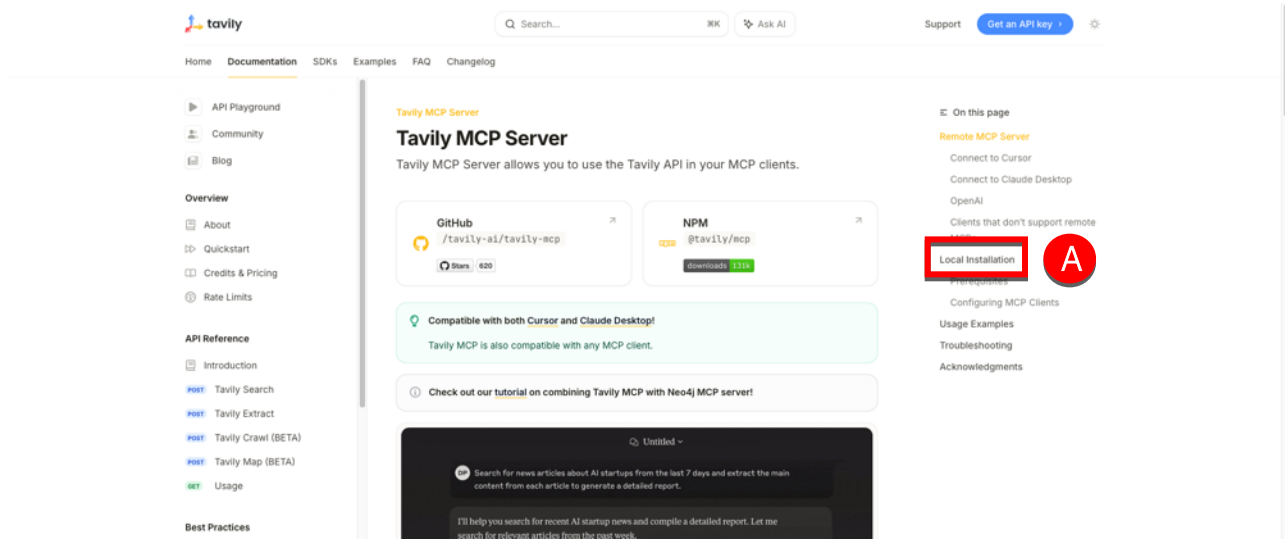
Get the MCP server installation command

Now that you have the API key, retrieve the command to install the MCP server from the MCP server documentation.

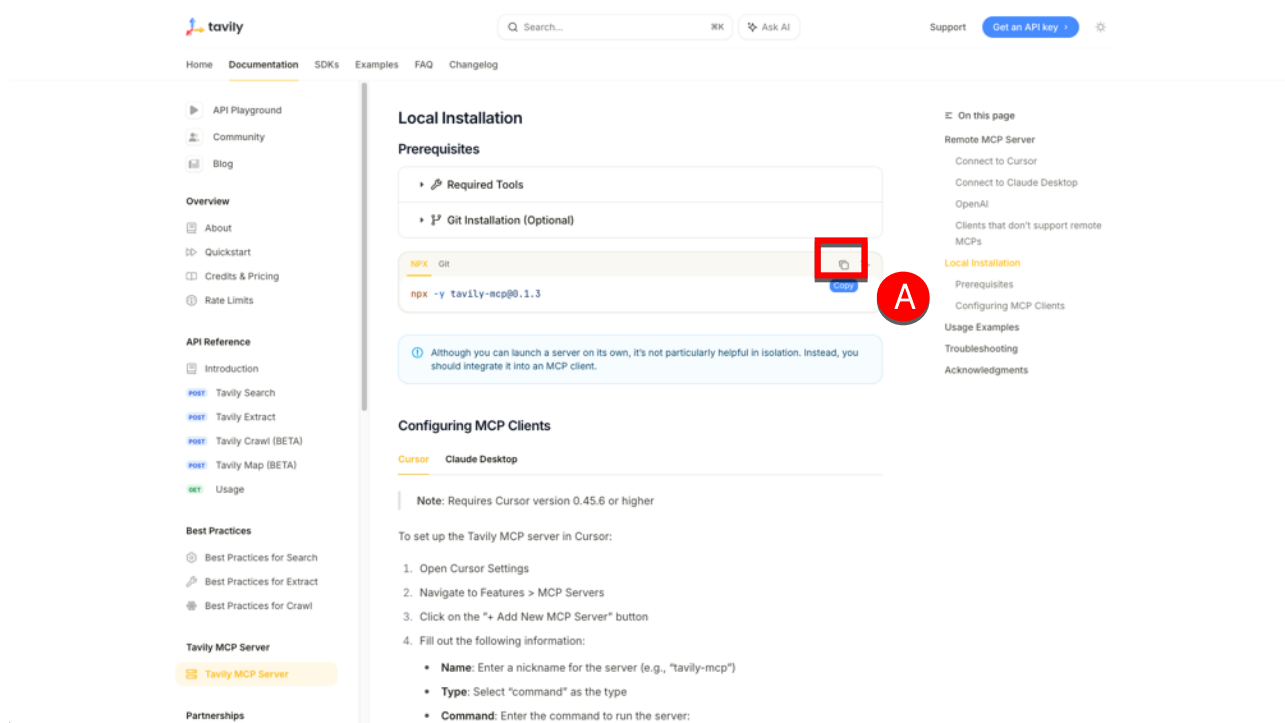
1. Click **Tavily MCP** (A).



2. Click Local installation (A).



3. Click the command line **Copy** icon (A) and copy it in your notes. You will use this command ('`npx -y tavily -mcp@#. #. #`') later to import the MCP tools inside your watsonx Orchestrate instance.



Reserve and prepare your environments

In this section you will reserve and configure two different environments (described in the architecture diagram) that you will use through the different labs in the watsonx Orchestrate L4.

Watsonx Orchestrate SaaS

During this L4 you will need a wxO SaaS trial to perform most of the labs contained in the new watsonx Orchestrate learning plan (September version). Follow these instructions to reserve and configure this SaaS environment.

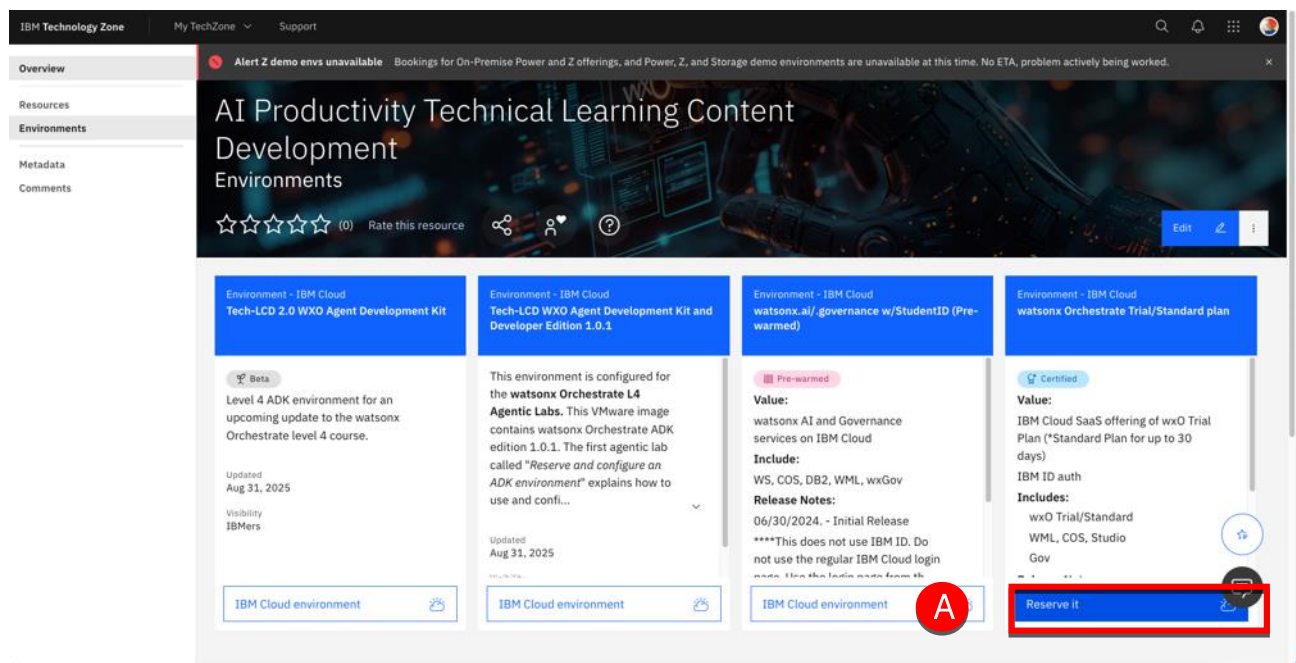
Reserve the environment

The environment can be reseed from IBM TechZone.

1. Click the following link to access the TechZone collection.

<https://techzone.ibm.com/collection/ai-productivity-technical-learning-content-development/environments>

2. Click **Reserve it** in the **watsonx Orchestrate Trial / Standard plan** tile (A).



Note: There are 2 SaaS trial tiles in the collection. Make sure you have clicked the one mentioning Account Vending in the title.

3. Click **Request an environment** (A).

IBM Technology Zone | My TechZone | Support

Create a reservation

AI Productivity Technical Learning Content Development

View collection

A Select an environment/infrastructure | Select a reservation type | Fill out your reservation | Complete

environment reservation options:

☒ Request an environment

☐ Schedule a workshop

Request multiple environments:

Leverage workshop manager to request multiple environments for your upcoming hands-on workshop and manage them all in one place. All workshop requests must be submitted 7 days before the intended workshop start date/time.

I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

Cookie Preferences

4. Select **Education** (A).

IBM Technology Zone | My TechZone | Support

Create a reservation

AI Productivity Technical Learning Content Development

View collection

Select an environment/infrastructure | Select a reservation type | **Fill out your reservation** | Complete

Name

watsonx Orchestrate Trial/Standard plan

Name this reservation. This will help identify it in your reservation list.

Purpose

A **Education**
Gaining experience with specific technology, product, or solution.

Demo
Deliver a client specific demonstration based on discovery with the client and aligns to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Lab Development
Build and configure environments to support the creation and testing of labs for the TechXchange event. Intended for lab developers during the development phase before final student delivery.

Pilot
Rapid co-creation build that proves IBM technologies can deliver business value to clients' end users. Serves as a foundation to build a production solution. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Test
Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can **NOT** be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Preferred Geography

Choose a preferred geography

I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

5. Type a Purpose description (A).

IBM Technology Zone | My TechZone | Support

Purpose

Demo
Deliver a client specific demonstration based on discovery with the client and aligns to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

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Test
Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can **NOT** be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Sales Opportunity number
Enter an opportunity number

Providing an [IBM Sales Cloud opportunity number](#), or a [Project Work ID](#) will allow you to extend your reservation date.

Purpose description
wxO L4 labs

What are you doing? Why do you need this? What are you trying to accomplish?

Preferred Geography
Choose a preferred geography

☐ I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

6. Select AMERICAS – us south region (A) in the Preferred Geography menu.

IBM Technology Zone | My TechZone | Support

Purpose

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Deliver a client specific demonstration based on discovery with the client and aligns to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

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Test
Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can **NOT** be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Sales Opportunity number
Enter an opportunity number

Providing an [IBM Sales Cloud opportunity number](#), or a [Project Work ID](#) will allow you to extend your reservation date.

Purpose description
wxO L4 labs

What are you doing? Why do you need this? What are you trying to accomplish?

Preferred Geography
Itz-watsonx - AMERICAS - us-south region - dal10 datacenter
Itz-watsonx - AP - jp-tok region - tok02 datacenter

☐ I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

7. Read and **Agree** the **IBM Technology Zone** conditions (A) then click **Submit** (B).

IBM Technology Zone

My TechZone

Support

Purpose

Demo

Deliver a client specific demonstration based on discovery with the client and aligns to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Lab Development

Build and configure environments to support the creation and testing of labs for the TechXchange event. Intended for lab developers during the development phase before final student delivery.

Test

Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can **NOT** be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Sales Opportunity number

Enter an opportunity number

Providing an [IBM Sales Cloud opportunity number](#), or a [Project Work ID](#) will allow you to extend your reservation date.

Purpose description

wxO L4 labs

What are you doing? Why do you need this? What are you trying to accomplish?

Preferred Geography

itz-watsonx- AMERICAS- us-south region- dal10 datacenter

Education

Gaining experience with specific technology, product, or solution.

Pilot

Rapid co-creation build that proves IBM technologies can deliver business value to clients' and users. Serves as a foundation to build a production solution. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

AI Productivity Technical Learning Content Development

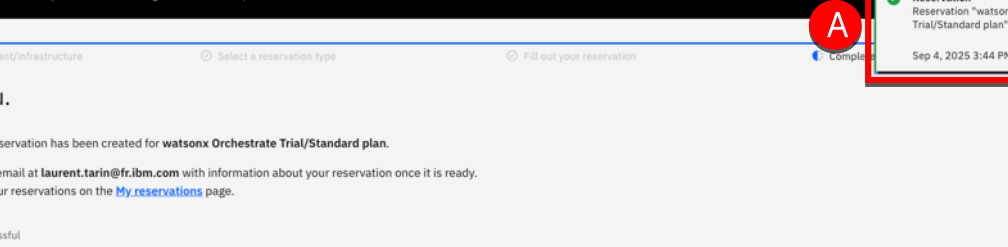
watsonx Orchestrate Trial/Standard plan

Duration Policy

Free to IBM Technology Zone's Terms & Conditions and End User Security Policies

Submit

8. Check that your reservation has been created (A) then wait for the “reservation is ready” email.



IBM Technology Zone My TechZone Support

Create a reservation

AI Productivity Technical Learning Content Development

Select an environment/infrastructure Select a reservation type Fill out your reservation

Thank you.

Your **ibmcloud-2** reservation has been created for **watsonx Orchestrate Trial/Standard plan**.

You will receive an email at **laurent.tarin@fr.ibm.com** with information about your reservation once it is ready.

You can manage your reservations on the [My reservations](#) page.

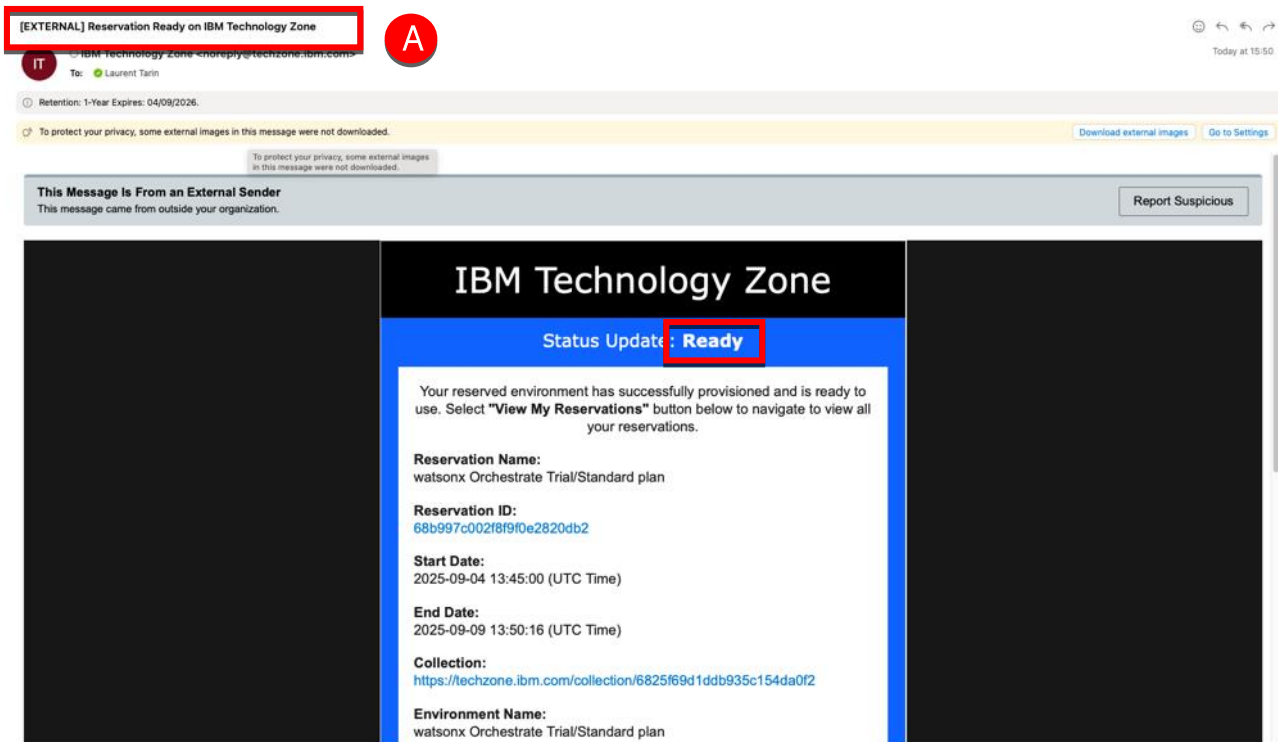
Submission successful

[My reservations](#)

Reservation "Reservation "watsonx Orchestrate Trial/Standard plan" created

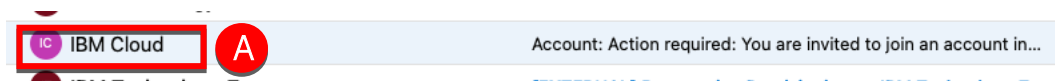
Sep 4, 2025 3:44 PM (+02:00)

9. Wait for the READY confirmation email from IBM Technology zone (A):



Join the IBM Account

Your watsonx Orchestrate instance is hosted on an IBM Cloud account. You will first need to join this account. If you have received another email from IBM Cloud (A) inviting you to join an account, open this email.



- The screenshot shows an email interface at the top with a subject line "Account: Action required: You are invited to join an account in IBM Cloud". The sender is "IBM Cloud <no-reply@cloud.ibm.com>" and the recipient is "Laurent Tarin". A retention notice states "Retention: 1-Year Expires: 25/08/2026". Below this is a yellow banner indicating that some external images were not downloaded for privacy reasons.

The main body of the email has a dark header with the "IBM Cloud" logo and the word "Account". The content includes:

 - "Action required: You are invited to join an account in IBM Cloud"
 - Source ID: 1756125020166
 - Type: Account
 - Update Time: 25 Aug 2025, 12:30 PM UTC
 - A link to "View notification (in subscribed accounts):"

The email continues with a greeting "Hi laurent.tarin@fr.ibm.com," followed by a red box labeled 'A' containing the text "account: itz-saas-332 (Account ID: 00772170c929450d8e000000000000f8e)". This is followed by a red box labeled 'B' containing the text "Join now." and "The link to join the account expires in 30 days."

The email concludes with "Thank you, IBM Cloud" and a contact link for "IBM Cloud support". At the bottom, there is a copyright notice: "© Copyright IBM Corporation 2014, 2025."

-
- IBM Cloud
- Catalog Cost estimator Docs
- ## Join IBM Cloud
- You're almost there. Complete your IBM Cloud account details so that you can get started.
- Account information

laurent.tarin@ftr.ibm.com

Personal information

Laurent Tarin

United States

Account notice

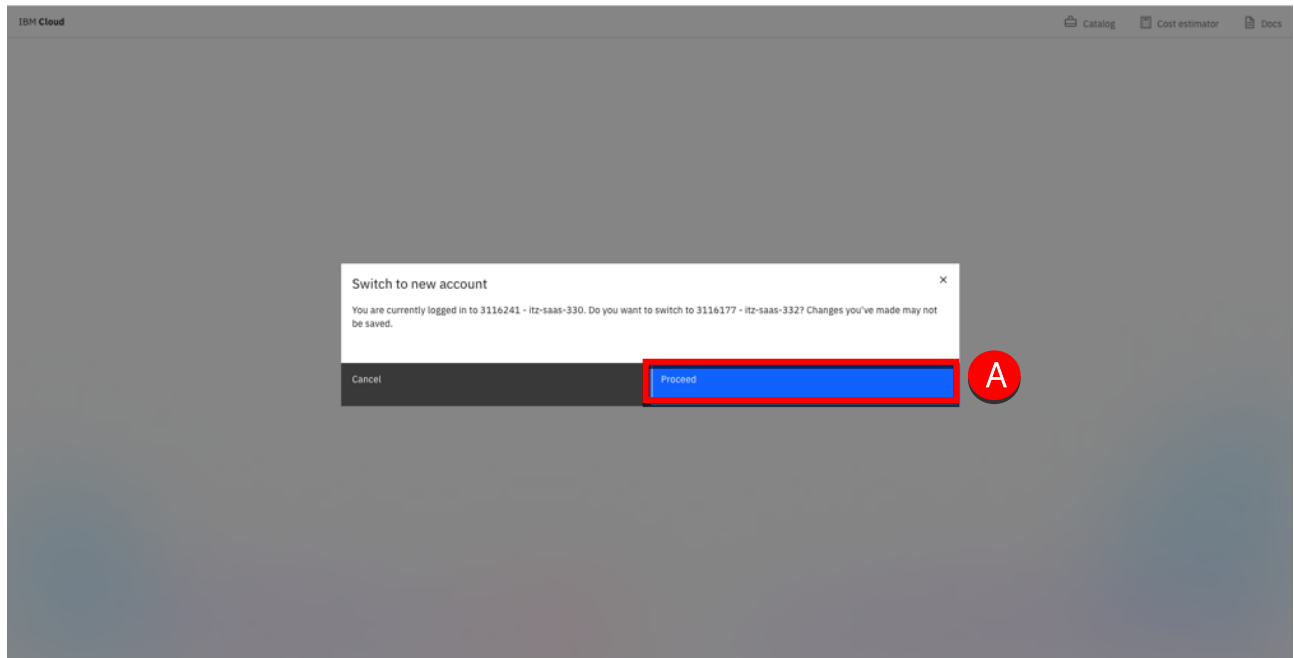
By providing my contact details, I acknowledge that IBM will keep me informed about products, services, and offers. More information on how IBM uses data can be found in the [IBM Privacy Statement](#). California residents, review the [California Supplemental Privacy Statement](#).

☒ I accept the product [Terms and Conditions](#) of this registration form.

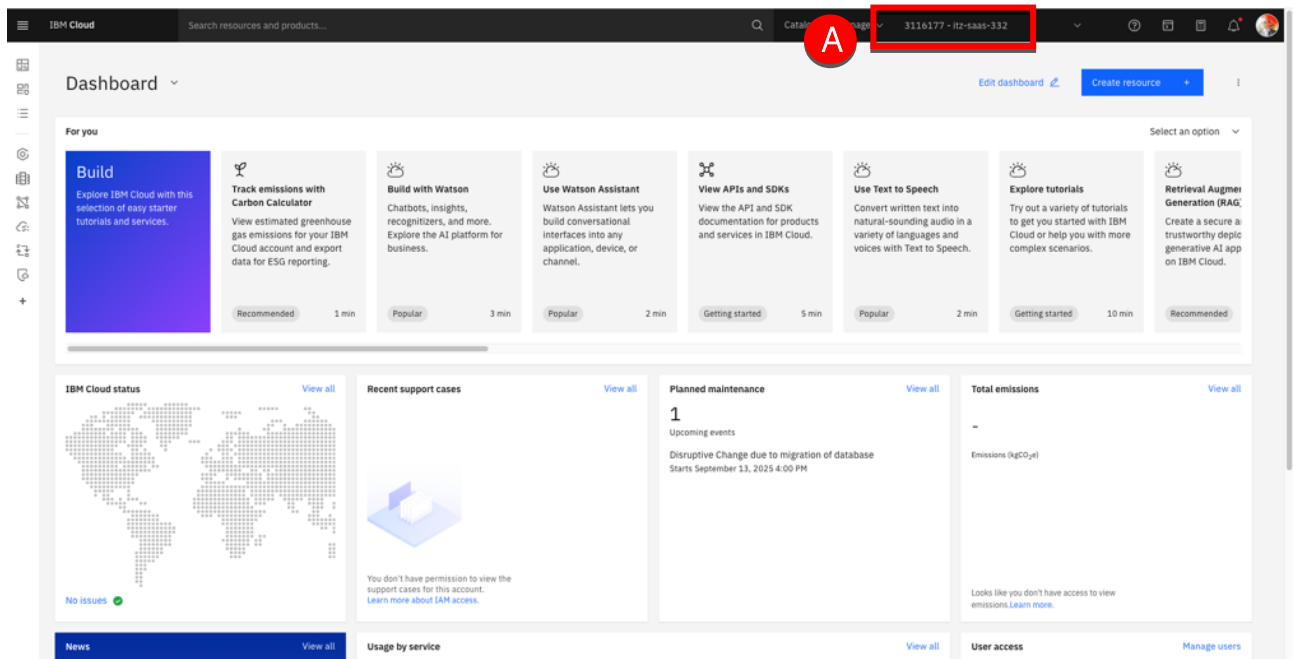
Join account
- Unlock the full IBM Cloud Catalog

Get started with everything you need to take your projects to the next level. Including always-free products like IBM Watson[®] APIs. They never expire and you can't be charged for them—ever.
- Cookie Preferences

5. Click **Proceed** to switch to the new account on IBM Cloud (A). Note, you will not see this message if this is your only IBM Cloud account.



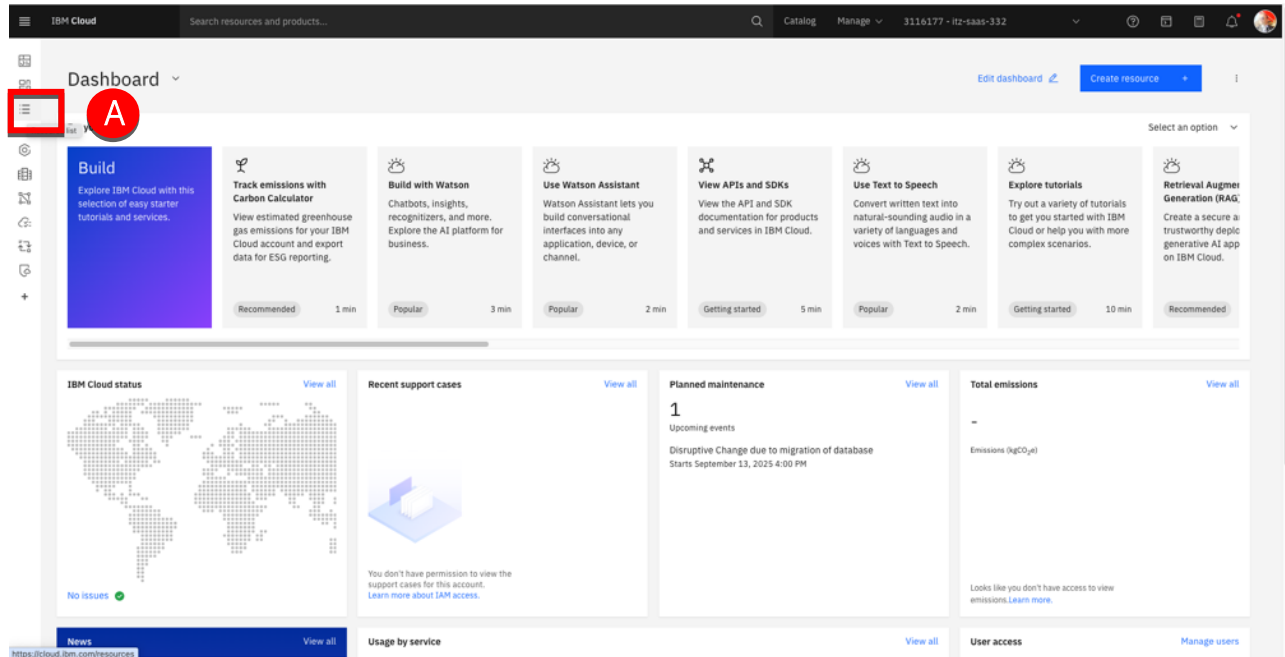
6. Make sure you are on the right cloud account as noted in item 6 in your lab credential notes. This is the IBM Cloud account named in the invitation email you received.



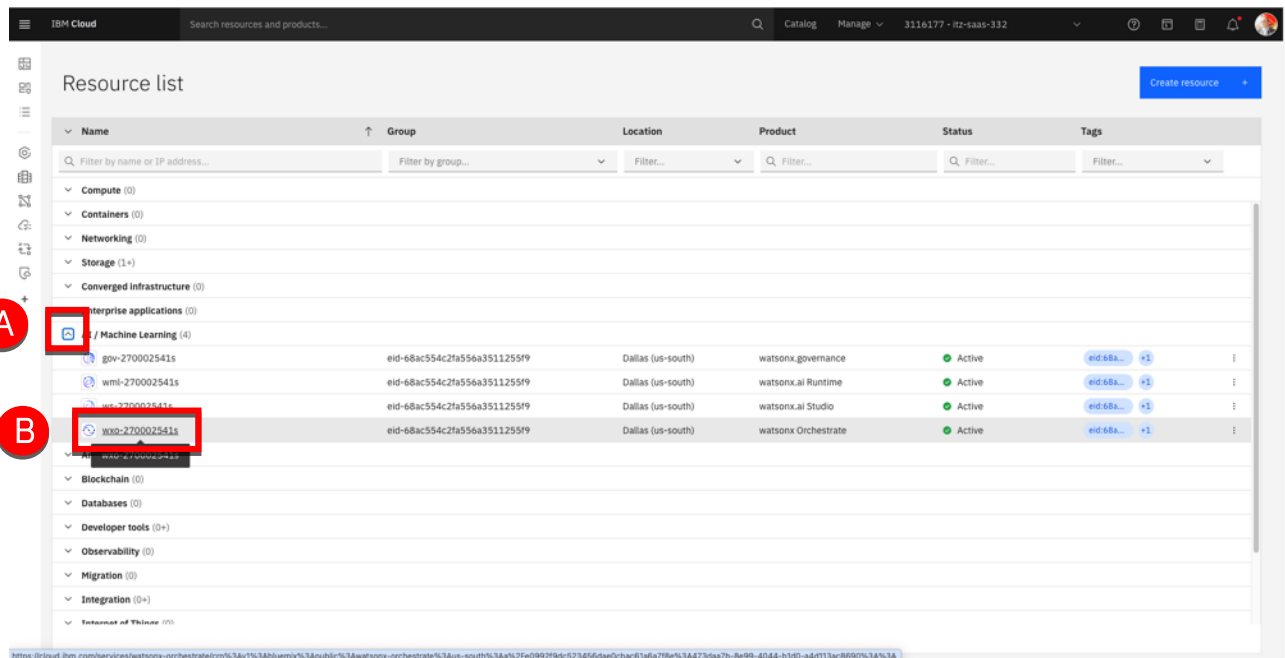
Get your instance URL

Next you will start your watsonx Orchestrate SaaS instance

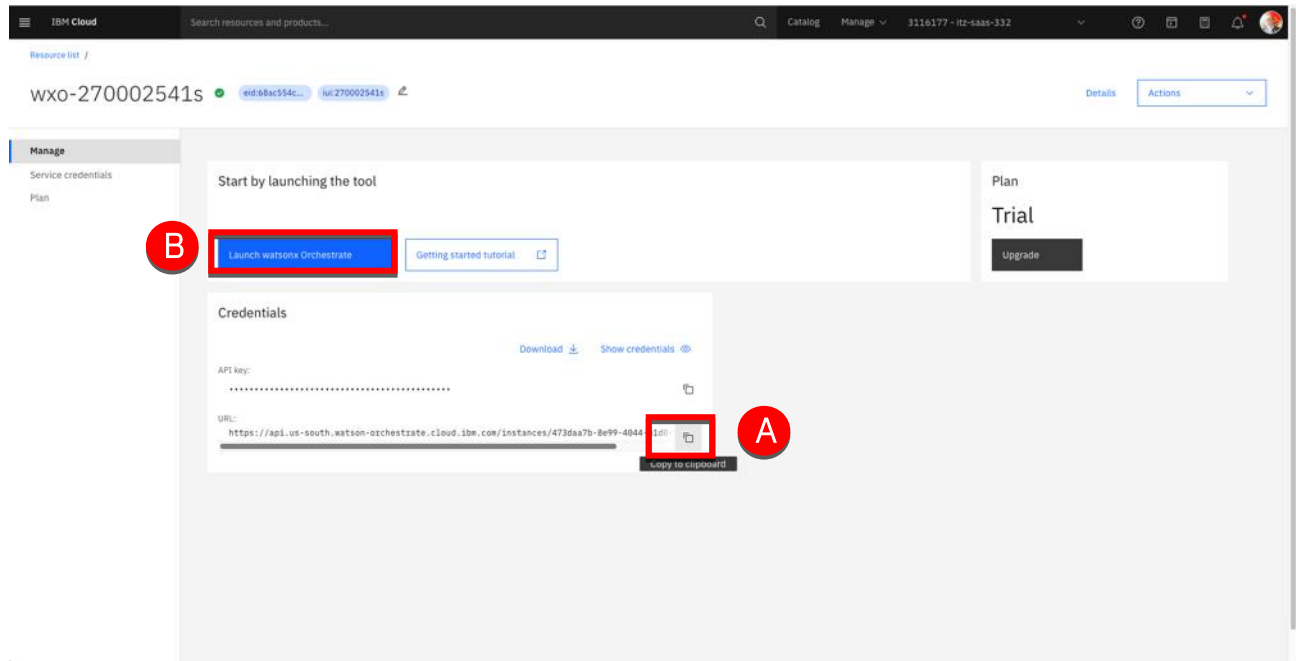
1. Click the Resource list (A).



2. Expand AI/Machine learning (A) then click watsonx Orchestrate (B).



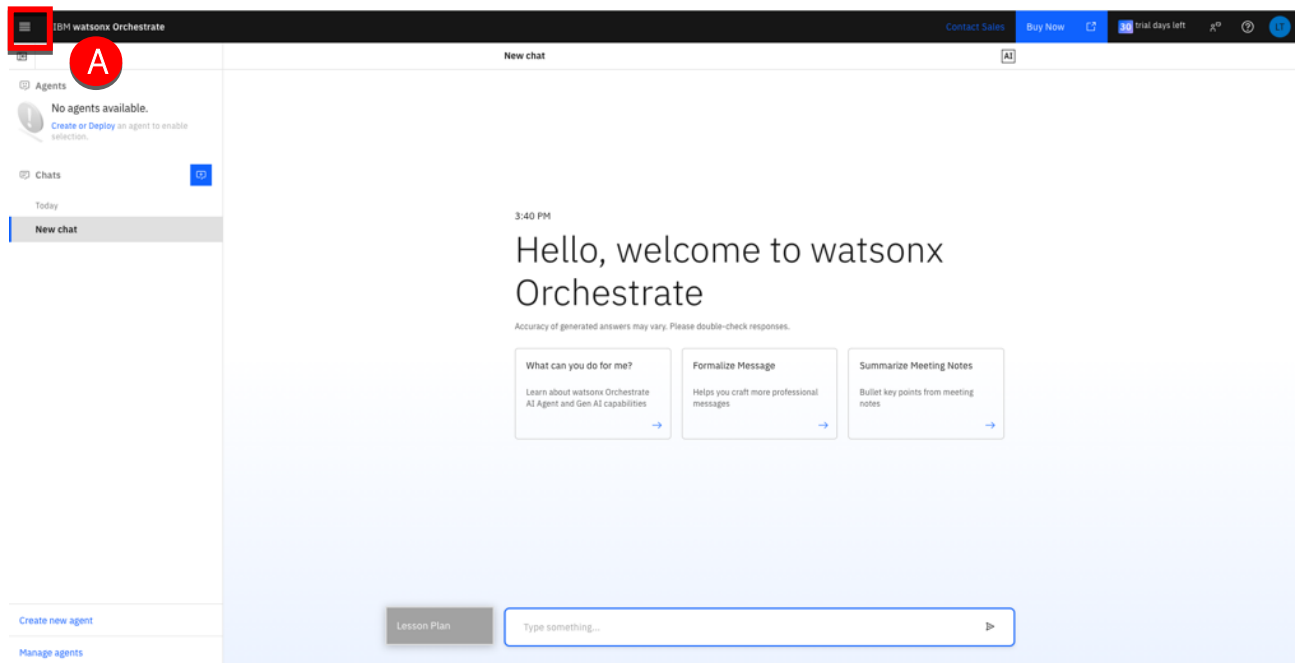
3. Click the copy to clipboard icon next to your instance URL (A) and copy this URL in your note (Item 7). Click **Launch watsonx Orchestrate** (B).



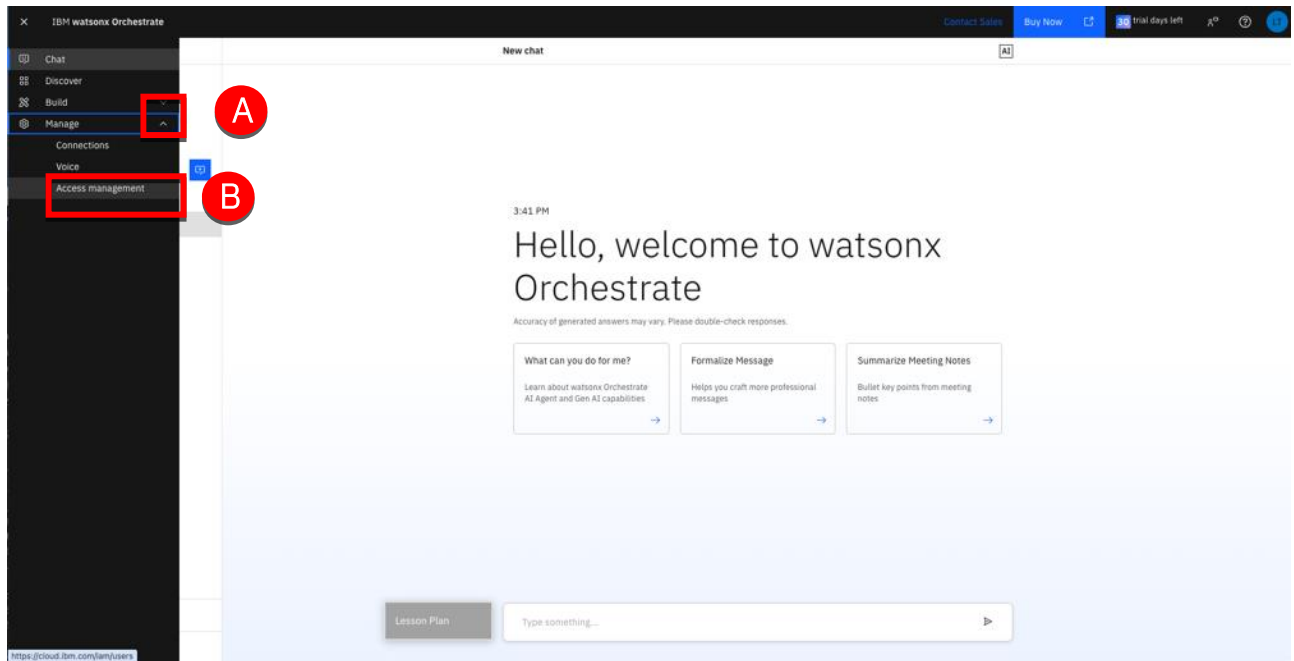
Create your API key

Next you will create a new API key (Item 8) to access the IBM Cloud account your watsonx Orchestrate SaaS is hosted on.

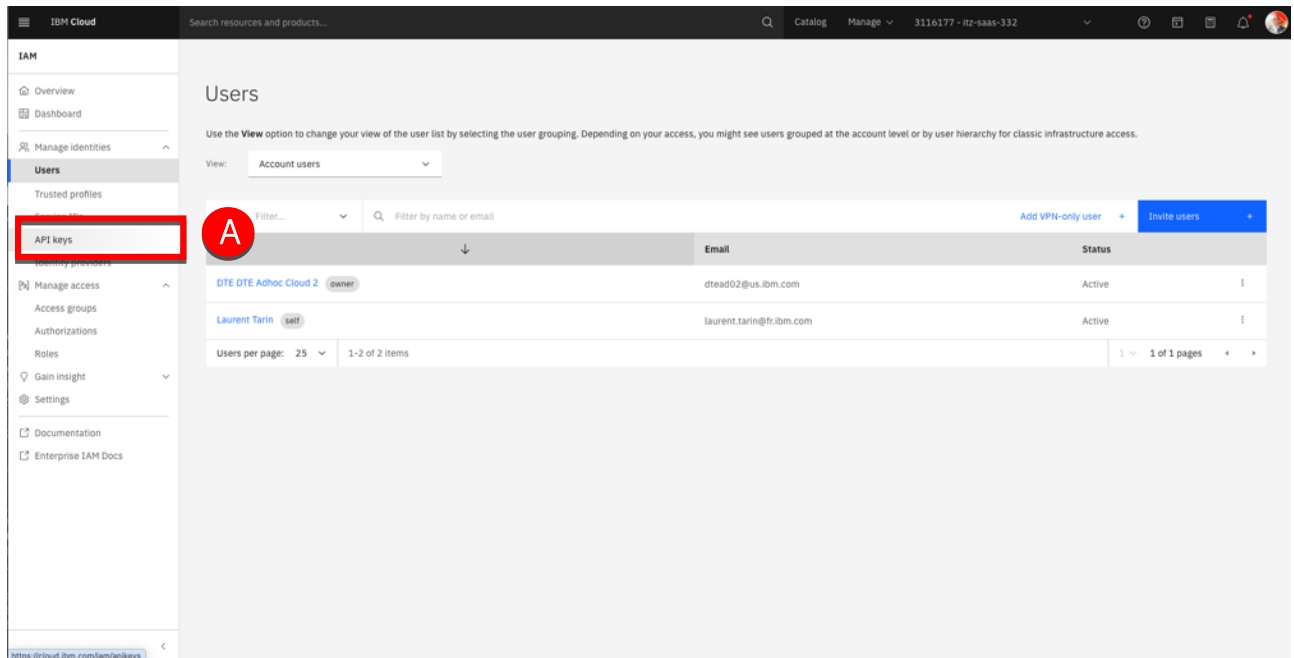
1. Click the main menu icon (A).



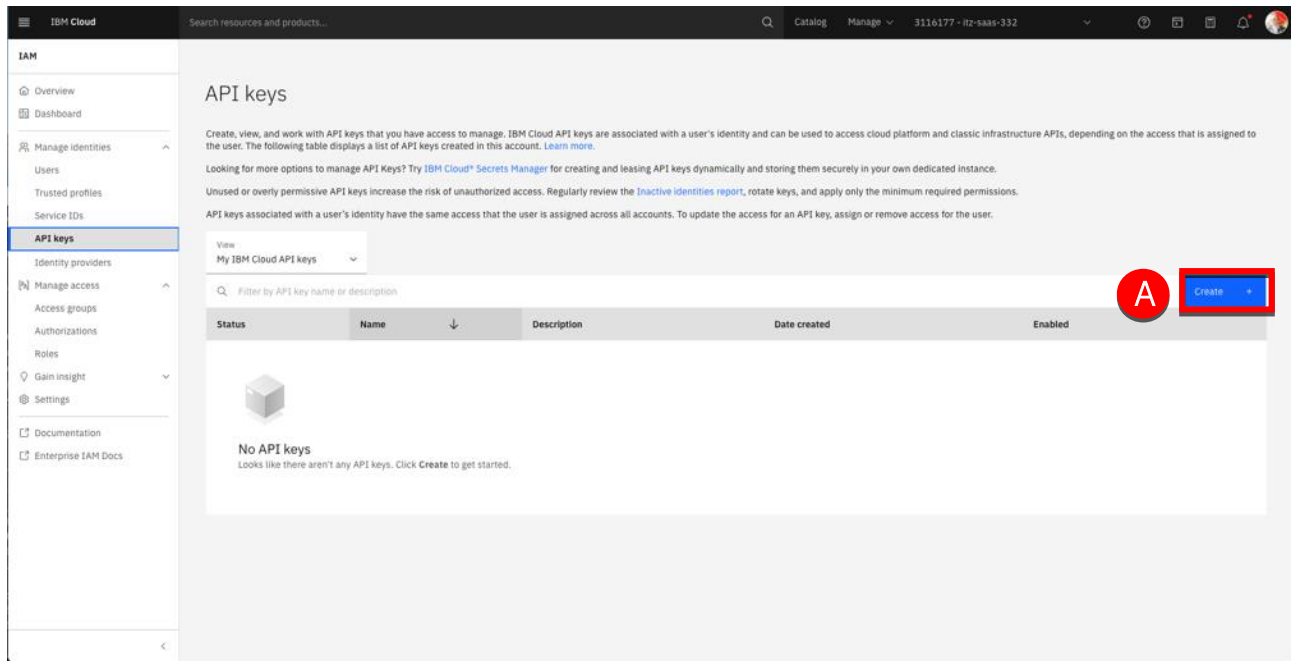
- Expand the **Manage** menu (A) then click **Access management** (B).



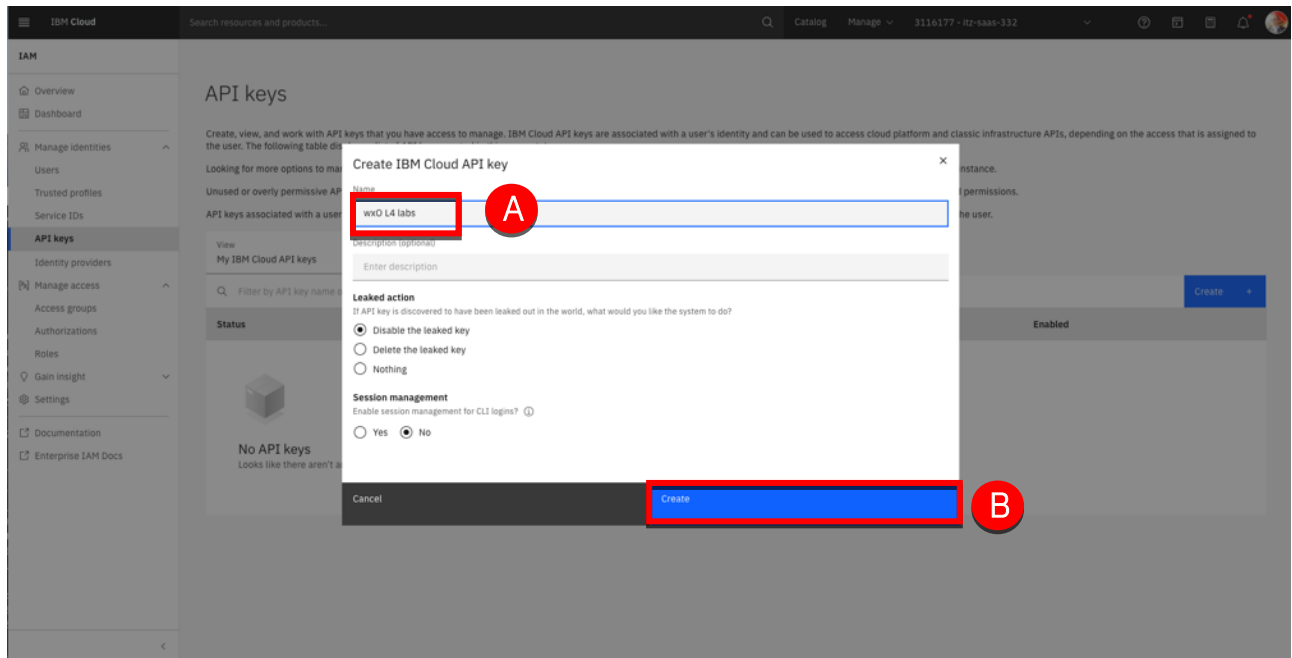
- Click **API Keys** (A).



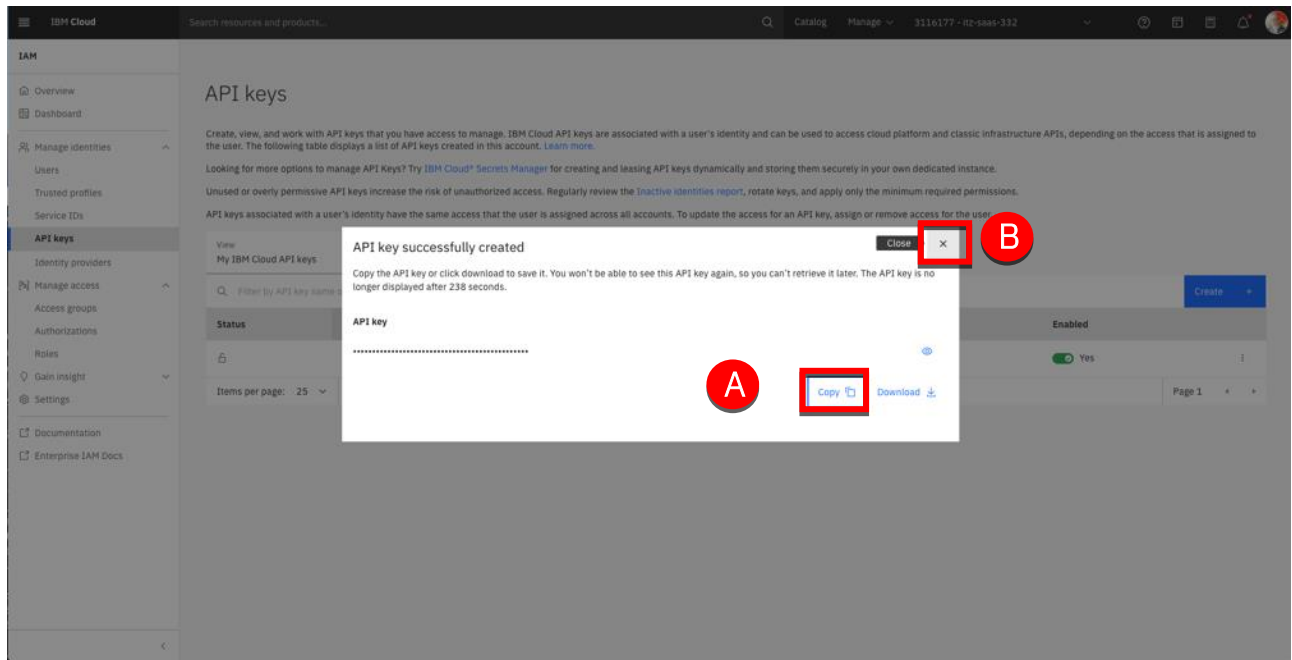
4. Click **Create +** (A).



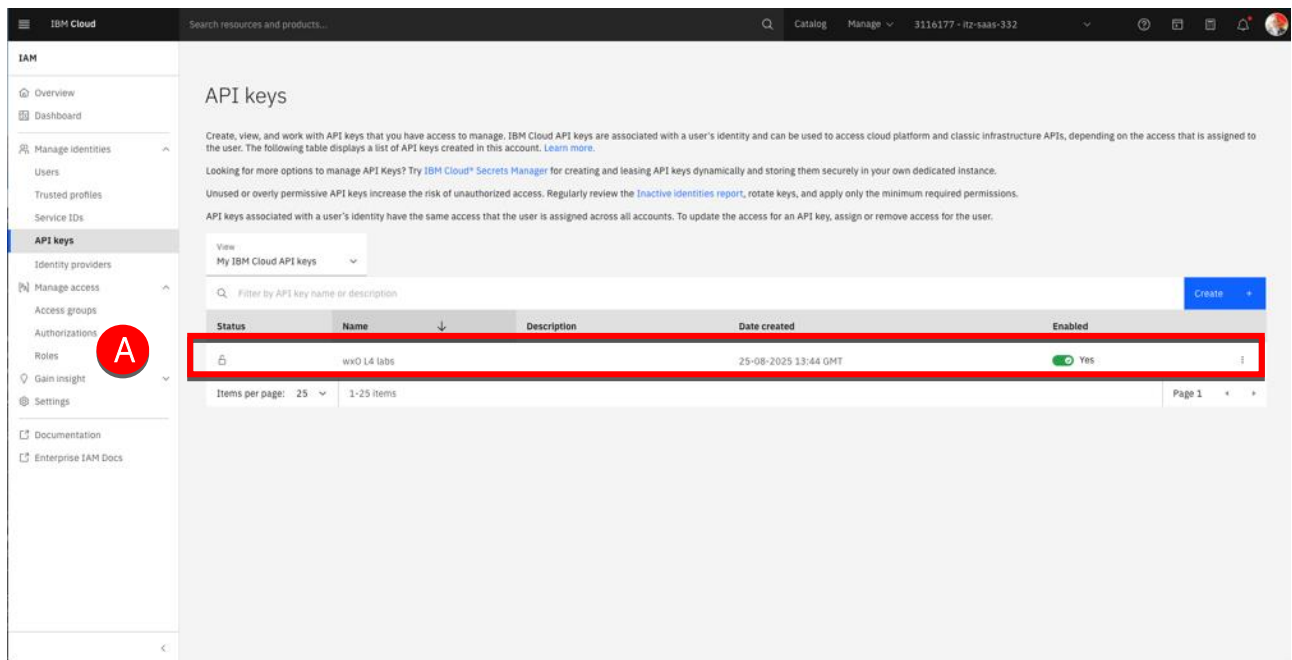
5. Enter any key name (A) then click **Create** (B).



- Click the **Copy** icon (A) then paste the API key value in item 8 from the lab credentials table you are referencing throughout this course. Click **Close** (B).



- Check that your key has been created (A) then close the browser tab.



Watsonx Orchestrate ADK

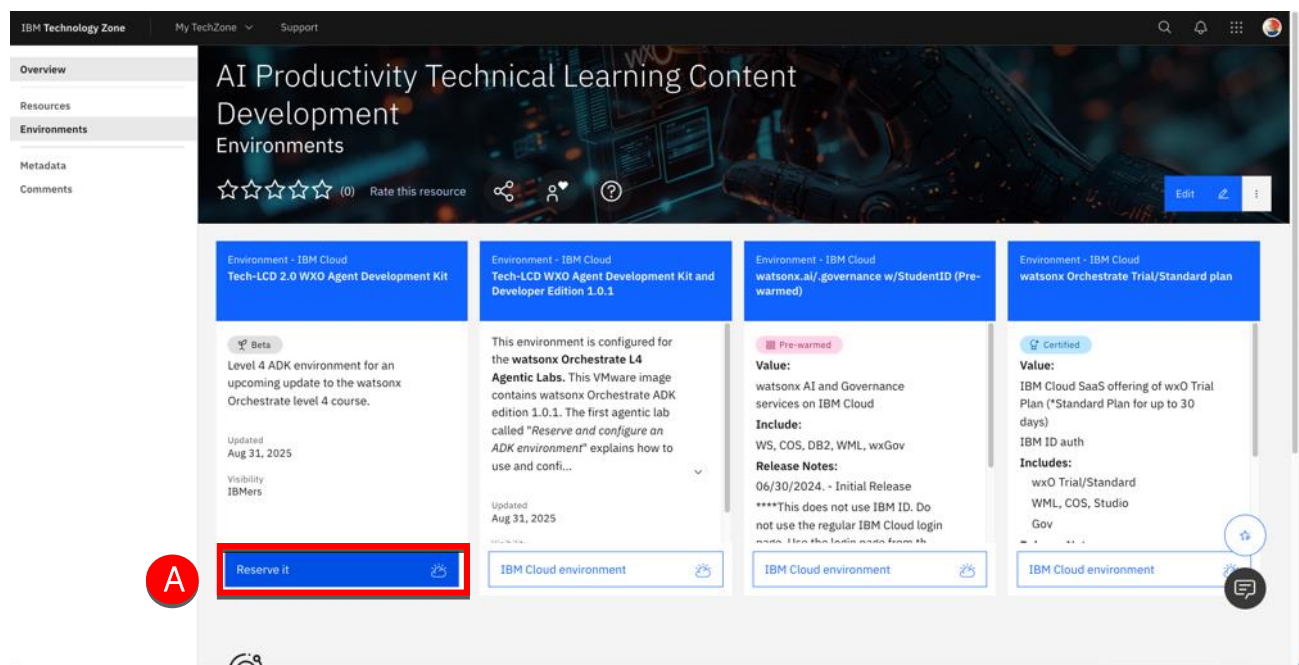
This section will provision a VM Ware image pre-configured with watsonx Orchestrate ADK and a local Developer Edition server.

Provision a VM image

1. Go to the TechZone collection here:

<https://techzone.ibm.com/collection/ai-productivity-technical-learning-content-development/environments>

2. Click **Reserve it** in the **Tech-LCD 2.0 W XO Agent Development Kit** tile (A).



Note: There may be more than one W XO Agent Development kit tile. Make sure you select the Tech-LCD 2.0 environment.

3. Check **Reserve** an environment (A).

IBM Technology Zone | My TechZone | Support

Create a reservation

Tech LCD - Environments Development Zone

View collection

Select a reservation type

Select your reservation type

Single environment reservation options:

☒ Request an environment

A

I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

4. Select **Education** (A).

IBM Technology Zone | My TechZone | Support

Create a reservation

Tech LCD - Environments Development Zone

View collection

Select a reservation type

Name

Tech-LCD 2.0 WXO Agent Development Kit

Name this reservation. This will help identify it in your reservation list.

Purpose

Demo

Deliver a client specific demonstration based on discovery with the client and aligns to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Education

Gaining experience with specific technology, product, or solution.

Lab Development

Build and configure environments to support the creation and testing of labs for the TechXchange event. Intended for lab developers during the development phase before final student delivery.

Pilot

Rapid co-creation build that proves IBM technologies can deliver business value to clients' end users. Serves as a foundation to build a production solution. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Test

Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can NOT be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Preferred Geography

Choose a preferred geography

I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

5. Type a Purpose description (A), Scroll down (B).

IBM Technology Zone | My TechZone | Support

Understand a customer specific demonstration based on discovery with the customer and align to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Lab Development
Build and configure environments to support the creation and testing of labs for the TechXchange event. Intended for lab developers during the development phase before final student delivery.

Pilot
Rapid co-creation build that proves IBM technologies can deliver business value to clients' end users. Serves as a foundation to build a production solution. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Test
Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can **NOT** be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Sales Opportunity number
Enter an opportunity number

Providing an [IBM Sales Cloud opportunity number](#), or a [Project Work ID](#) will allow you to extend your reservation date.

Purpose description
wxO L4 lab **A**

What are you doing? Why do you need this? What are you trying to accomplish?

Preferred Geography
Choose a preferred geography

Notes
Enter any notes your would like to attach to this reservation

☐ I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

B

6. Expand the Select a geography menu.

IBM Technology Zone | My TechZone | Support

Alert Known Issue: TechZone user experiencing failed reservations and stuck in provisioning The TechZone team is actively investigating an issue with IBM Cloud S3 storage. This is currently impacting all provisioning.

architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Pilot
Rapid co-creation build that proves IBM technologies can deliver business value to clients' end users. Serves as a foundation to build a production solution. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Test
Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can **NOT** be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Sales Opportunity number
Enter an opportunity number

Providing an [IBM Sales Cloud opportunity number](#), or a [Project Work ID](#) will allow you to extend your reservation date.

Purpose description
wxO ADK L4 lab

What are you doing? Why do you need this? What are you trying to accomplish?

Preferred Geography
Choose a preferred geography **A**

Notes
Enter any notes your would like to attach to this reservation

☐ I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

7. Select an available geography (A).

IBM Technology Zone | My TechZone | Support

Discover a time-specific demonstration based on discovery, enablement and rights to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Lab Development
Build and configure environments to support the creation and testing of labs for the TechXchange event. Intended for lab developers during the development phase before final student delivery.

Pilot
Rapid co-creation build that proves IBM technologies can deliver business value to clients' end users. Serves as a foundation to build a production solution. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Test
Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can **NOT** be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Sales Opportunity number
Enter an opportunity number

Providing an [IBM Sales Cloud opportunity number](#), or a [Project Work ID](#) will allow you to extend your reservation date.

Purpose description
wxO L4 labs

What are you doing? Why do you need this? What are you trying to accomplish?

Preferred Geography
ITZVMWARE-ITZEU3-LON02 - EUROPE - eu-gb region - lon02 datacenter

Start date and time
Select a date
08/25/2025 04:30 PM | Europe/Paris

Tech LCD - Environments Development Zone
Tech-LCD 2.0 WxO Agent Development Kit
Duration Policy

I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

Submit

8. Read and agree with the IBM Terms & conditions (A), click Submit (B).

IBM Technology Zone | My TechZone | Support

Discover a time-specific demonstration based on discovery, enablement and rights to the identified architecture. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Lab Development
Build and configure environments to support the creation and testing of labs for the TechXchange event. Intended for lab developers during the development phase before final student delivery.

Pilot
Rapid co-creation build that proves IBM technologies can deliver business value to clients' end users. Serves as a foundation to build a production solution. Automatically captures a Technical Sales Activity in IBM Sales Cloud on the Opportunity code provided.

Test
Need to test a specific function, configuration, or customization.

Please ensure to select the correct purpose as this can **NOT** be updated or changed after this reservation has been created. Review the [Reservation Duration Policy](#) to understand default durations allowed for specific infrastructures based on purpose.

Sales Opportunity number
Enter an opportunity number

Providing an [IBM Sales Cloud opportunity number](#), or a [Project Work ID](#) will allow you to extend your reservation date.

Purpose description
wxO L4 labs

What are you doing? Why do you need this? What are you trying to accomplish?

Preferred Geography
ITZVMWARE-ITZEU3-LON02 - EUROPE - eu-gb region - lon02 datacenter

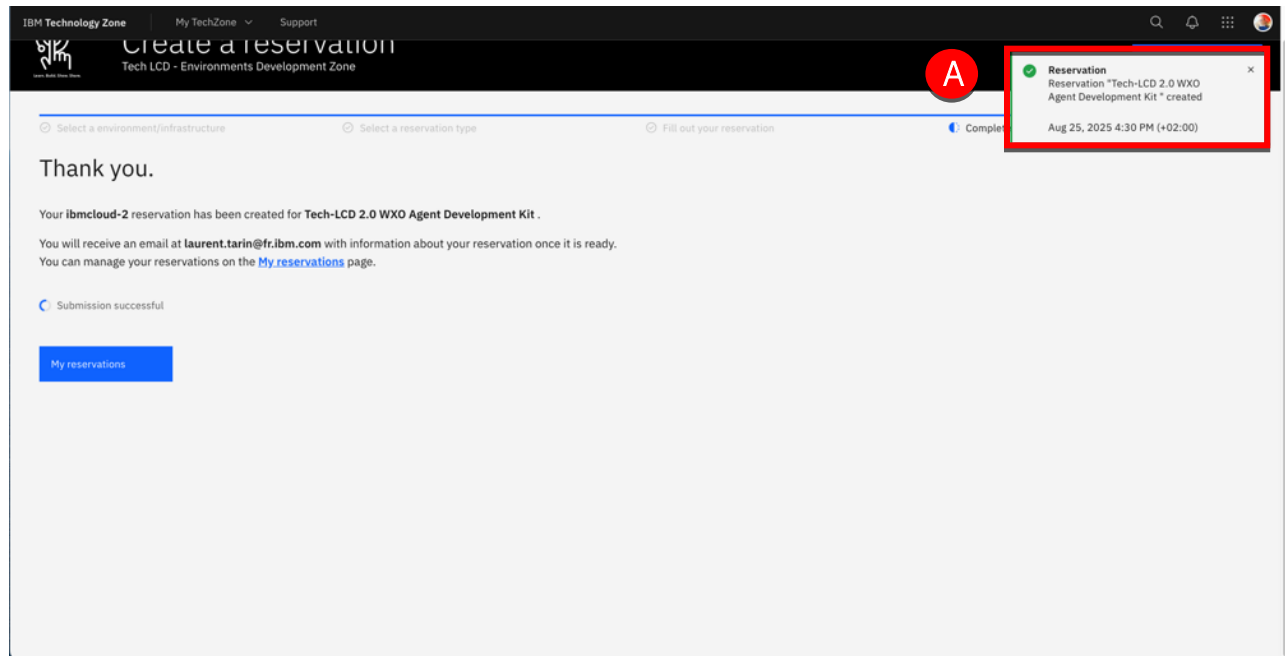
Start date and time
Select a date
08/25/2025 04:30 PM | Europe/Paris

Tech LCD - Environments Development Zone
Tech-LCD 2.0 WxO Agent Development Kit
Duration Policy

☒ I agree to IBM Technology Zone's [Terms & Conditions](#) and [End User Security Policies](#)

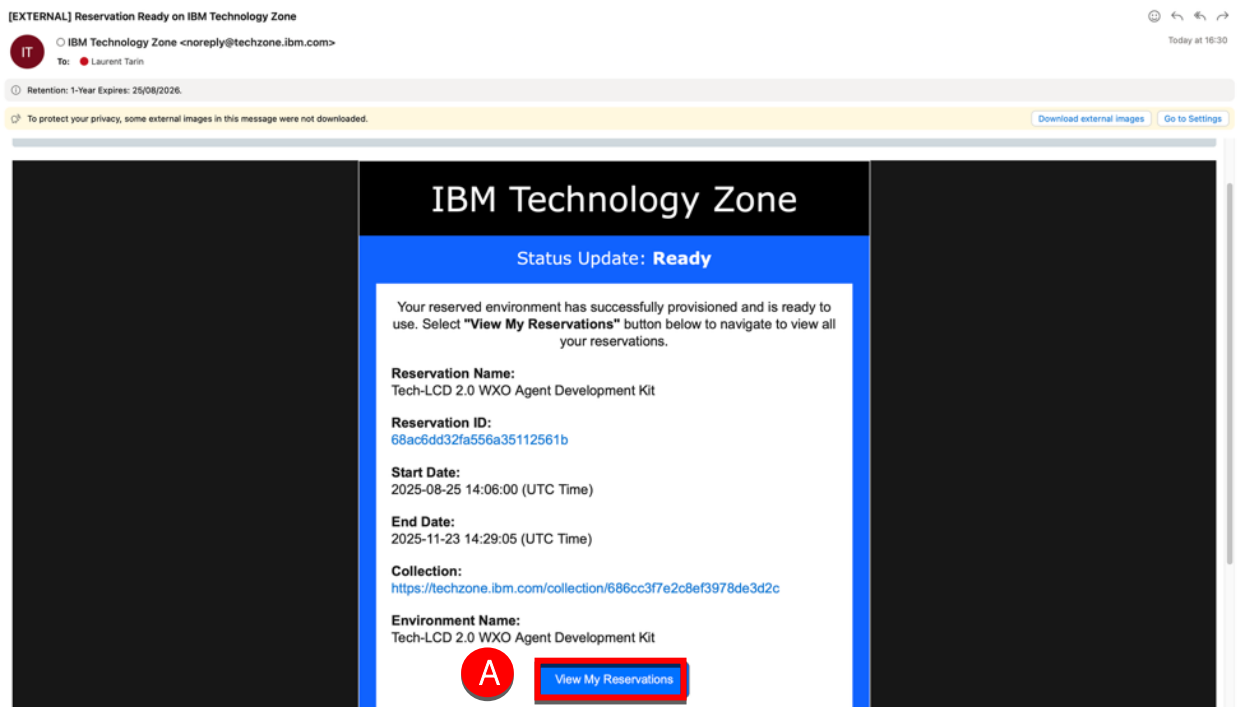
Submit

9. Ensure your reservation submission is correct (A) then wait for the status Ready email.

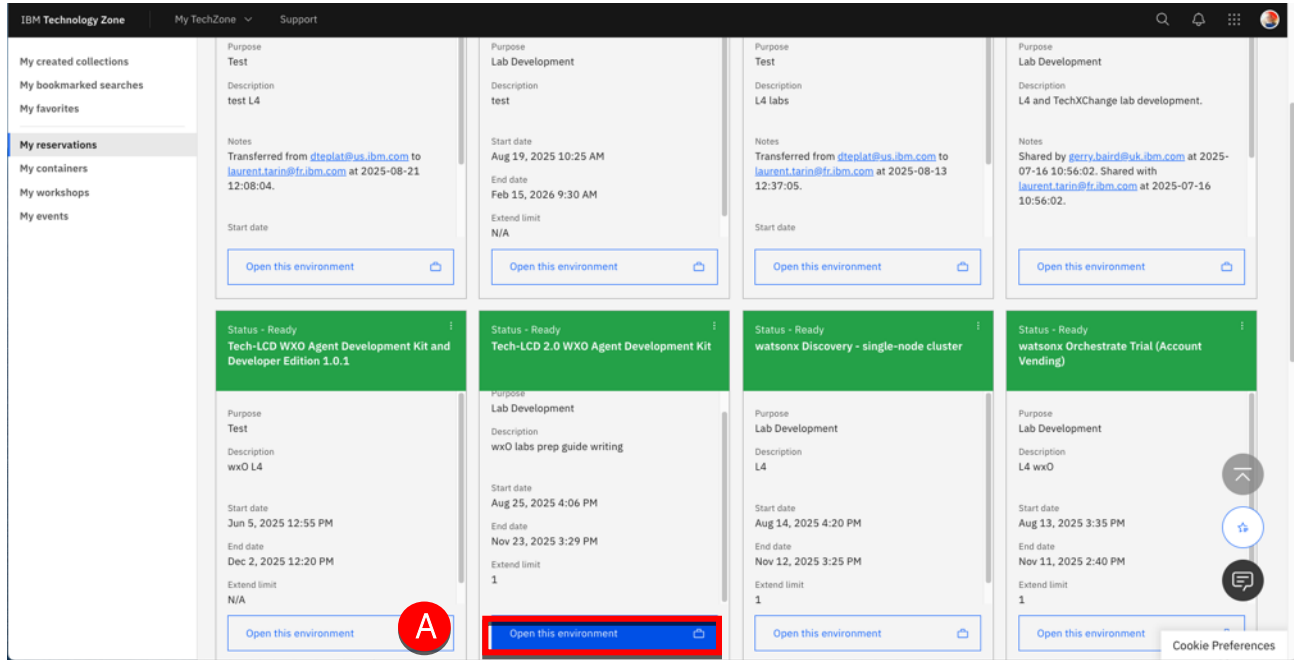


Get your credentials

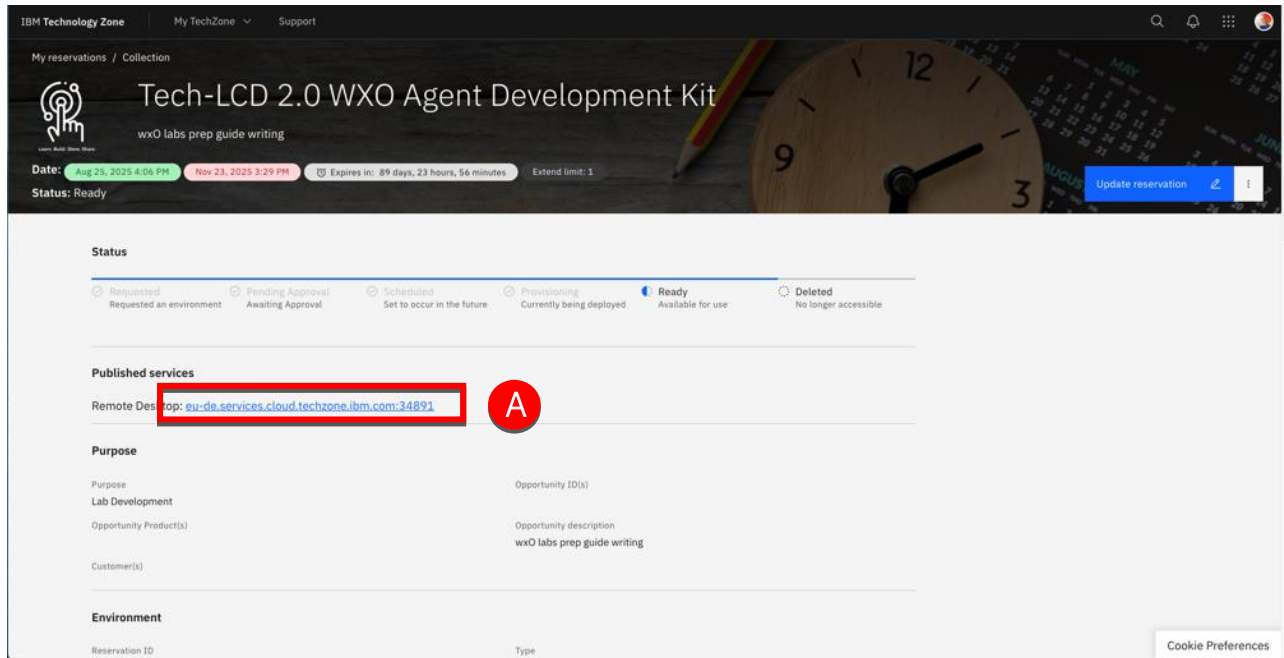
1. When the notification email arrives, open it and click **View my reservation** (A).



- Click **Open this environment** in the **Tech-LCD 2.0 WXO Agent Development Kit** tile (A).



- Copy the **Remote Desktop** URL in your note (Item 3).

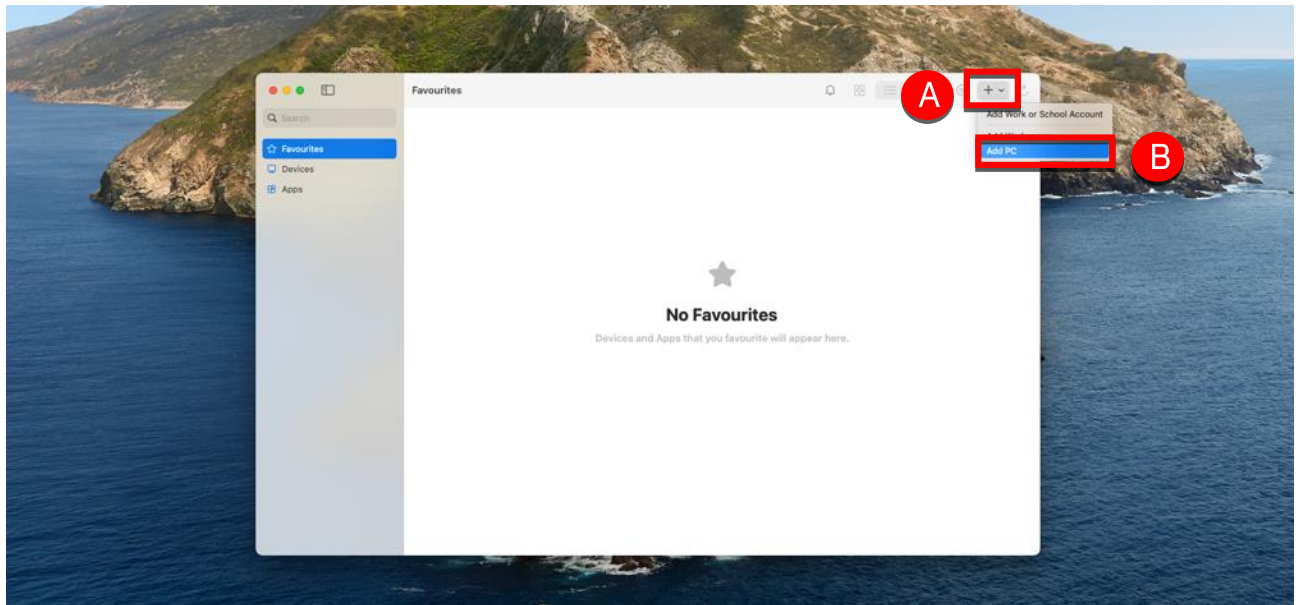


Access your image in Windows App (MacOS Users)

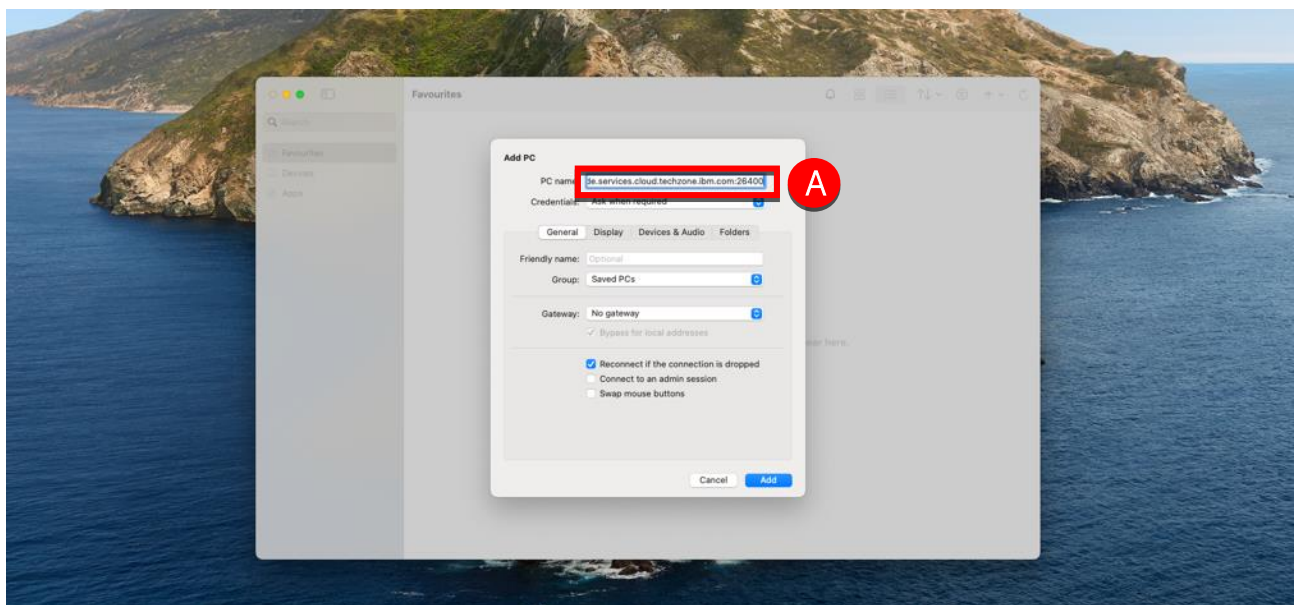
If you are a MS Windows user, you can skip this section and go to [Access your image in Remote Desktop App \(Windows Users\)](#). If you will be connecting to the remote machine using a Mac you will configure

Windows App to access the VMWare image containing your pre-configured watsonx Orchestrate Developer Edition environment. If you are using Windows, please skip to the next section.

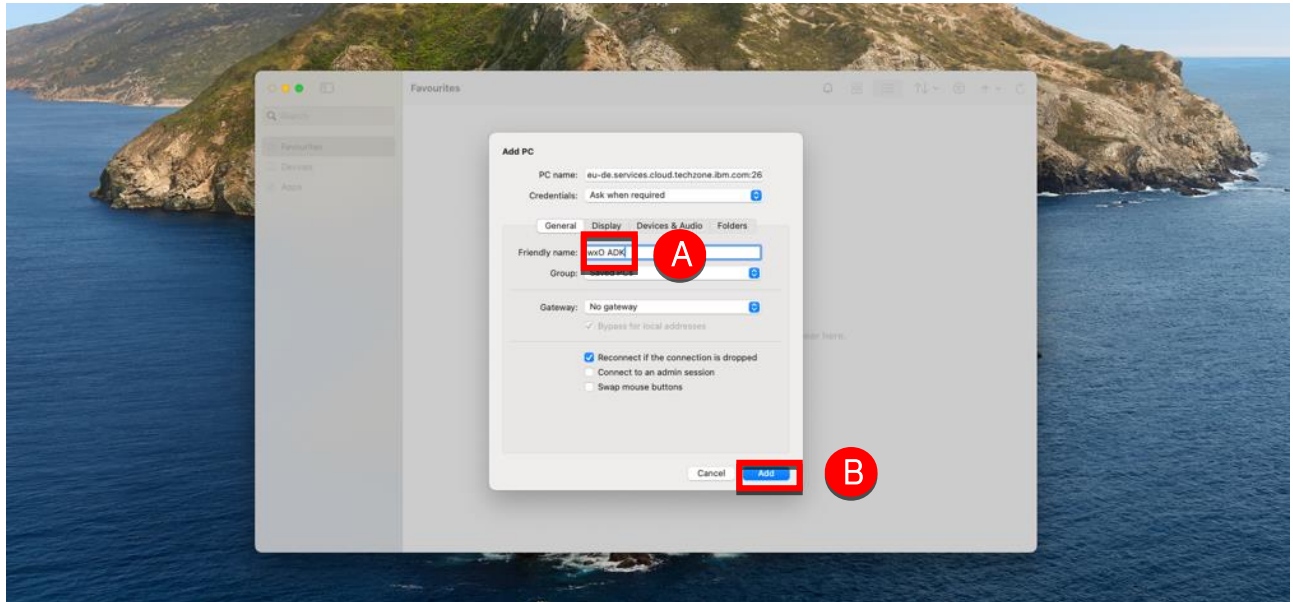
1. Start the Windows App application on your Mac.
2. Click + (A) then select **Add PC** (B).



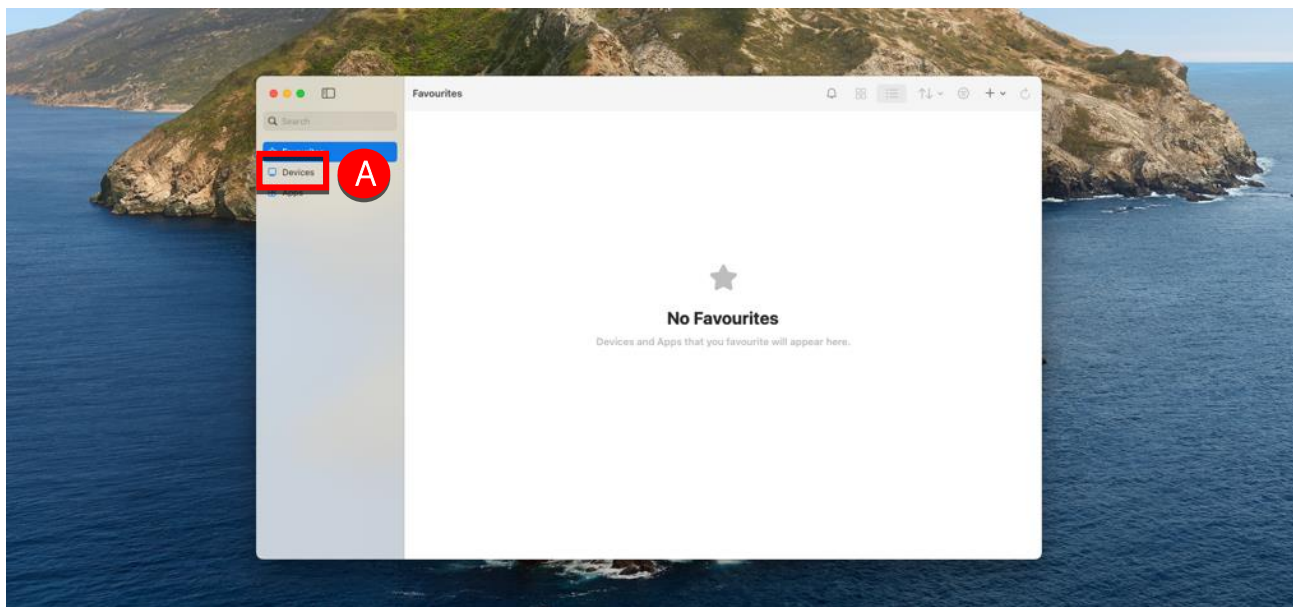
3. Copy your Remote Desktop URL from your lab credentials (item 3) into the **PC name** (A).



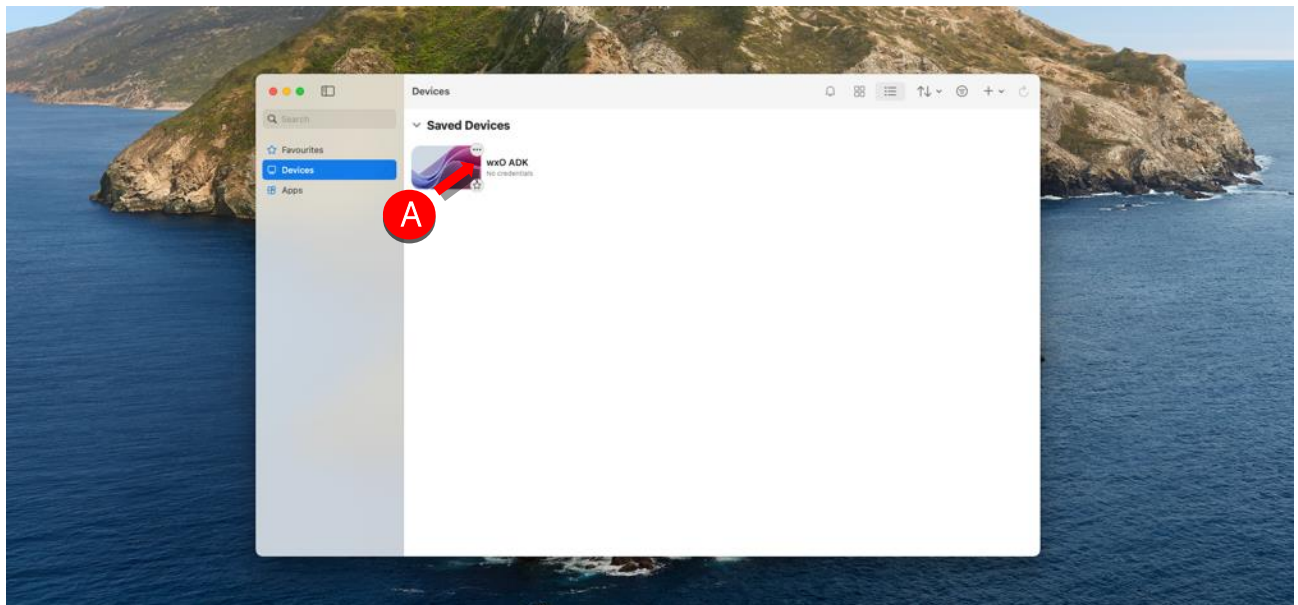
4. Enter “wx0 ADK” into the **Friendly name** field (A). Click **Add** (B).



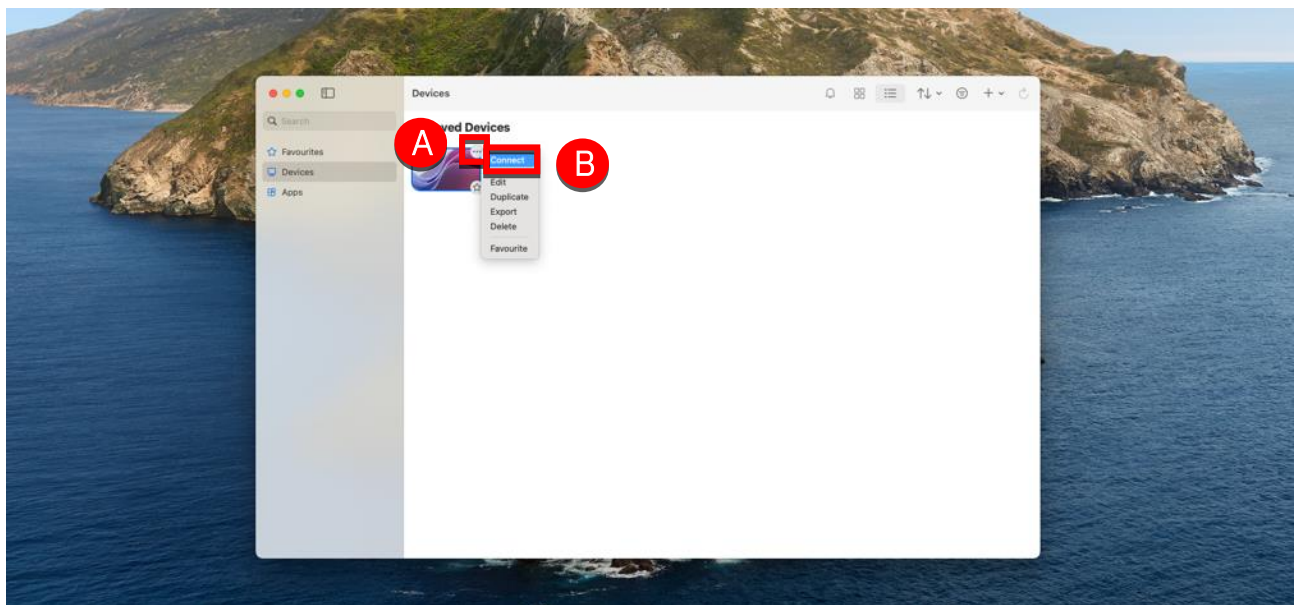
5. Click the **Devices** menu (A).



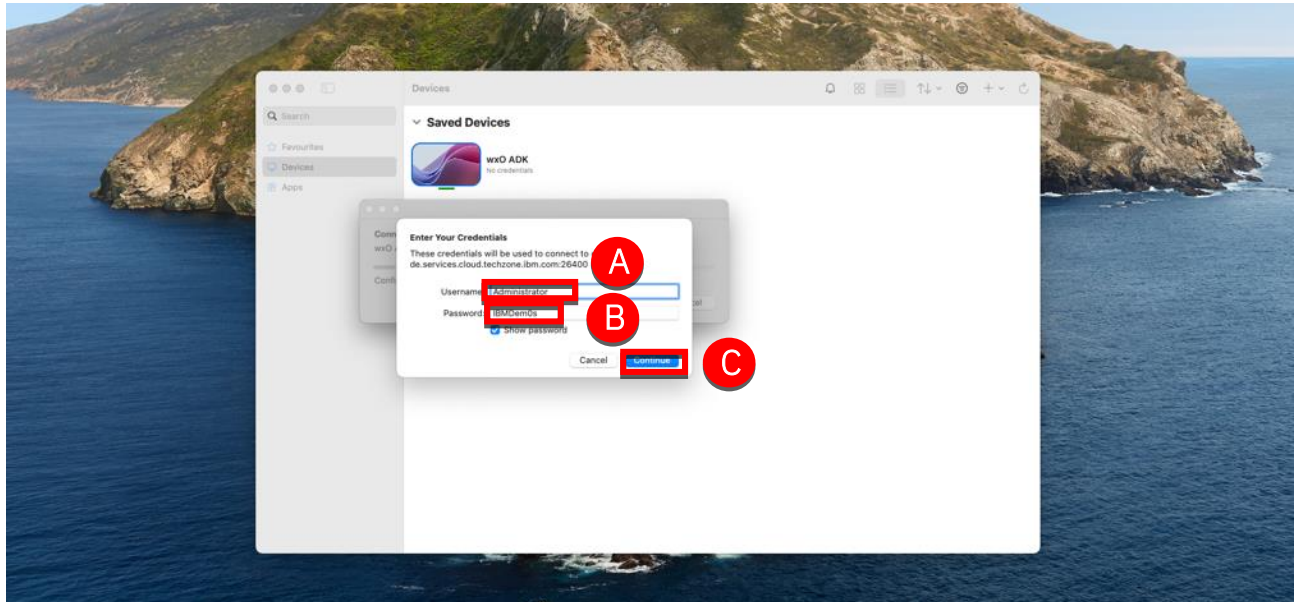
6. Place the cursor over your added PC to display the contextual menu (A).



7. Click the ... menu (A), then select **Connect** (B).

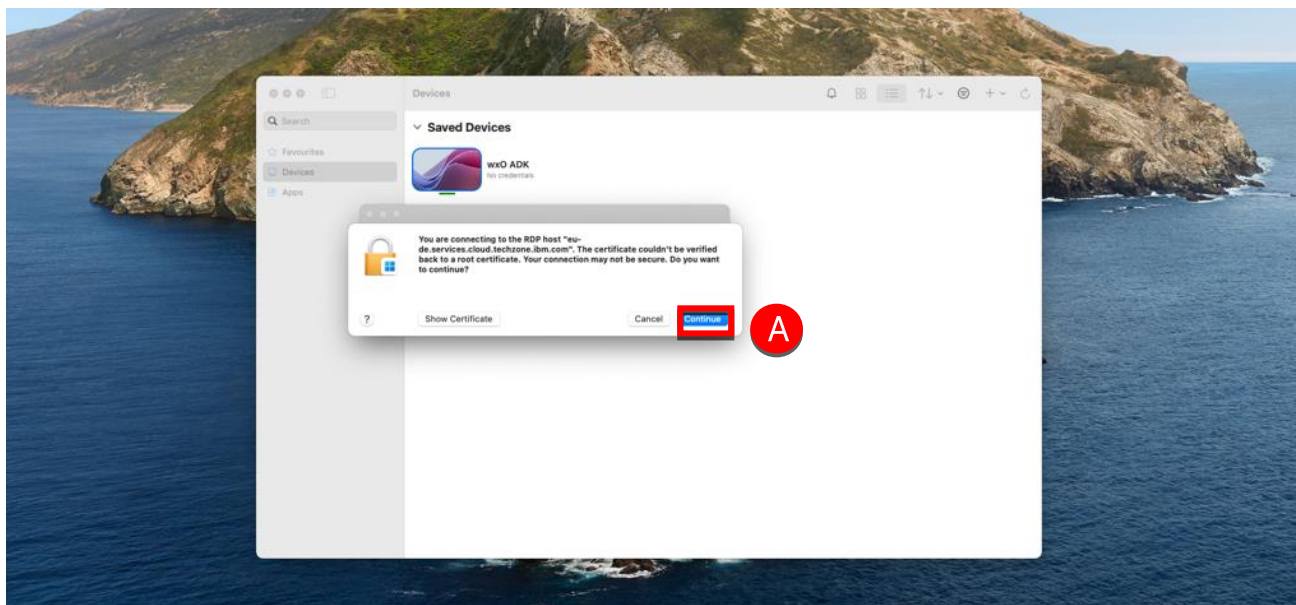


8. Type “Administrator” into **Username** (A). Type “IBMDemo0s” in **Password** (B). Click **Continue** (C).



Note: Some windows clients may need the “\” preceding “Administrator” (\Administrator) to indicate that this is an account on a remote machine, not on your local machine.

9. Click **Continue** (A).

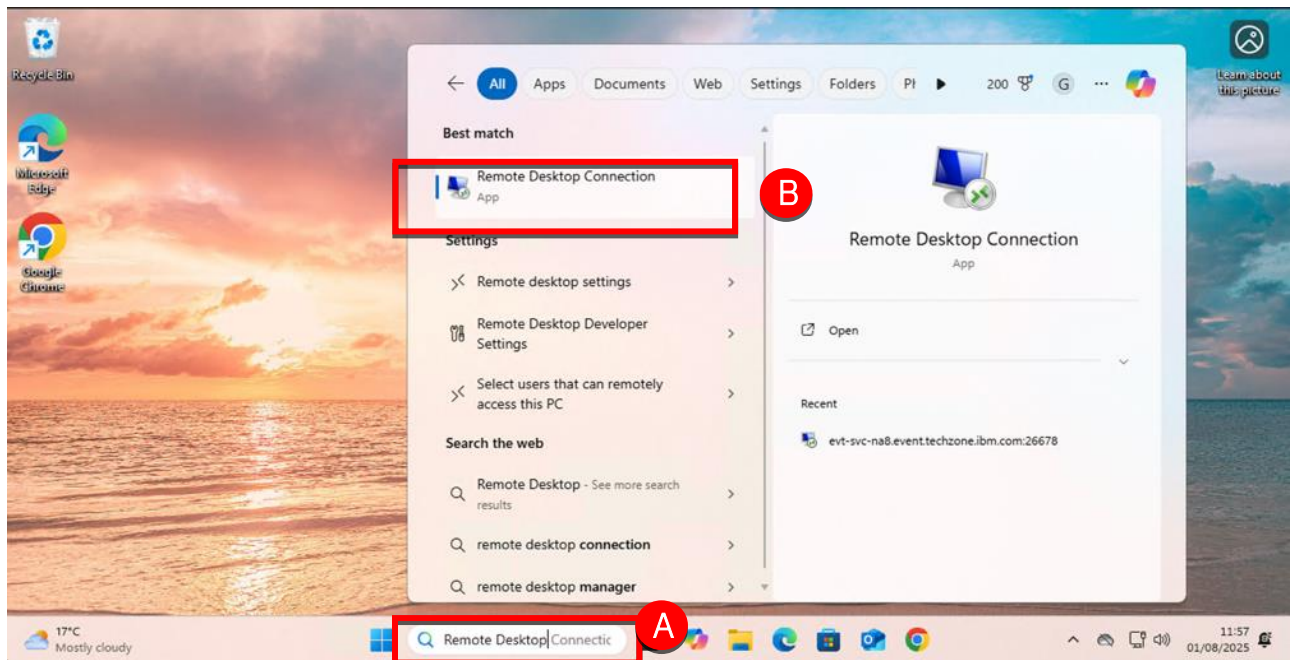


Access your image in Remote Desktop App (Windows Users)

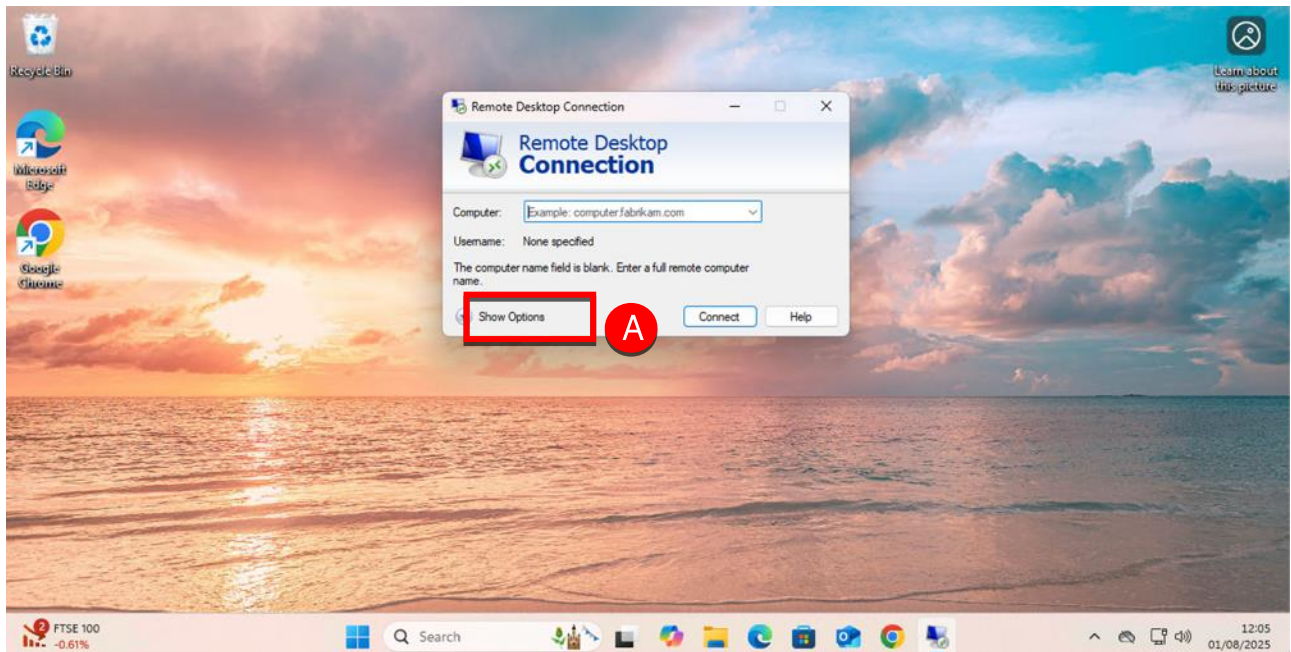
Skip this step if you have already connected to the remote machine from your Mac workstation. This section is specifically for Windows users connecting a Windows workstation to the remote machine.

If you will be connecting to the remote machine using a Windows workstation you will configure Remote Desktop Connection to access the VMWare image containing your pre-configured watsonx Orchestrate Developer Edition environment.

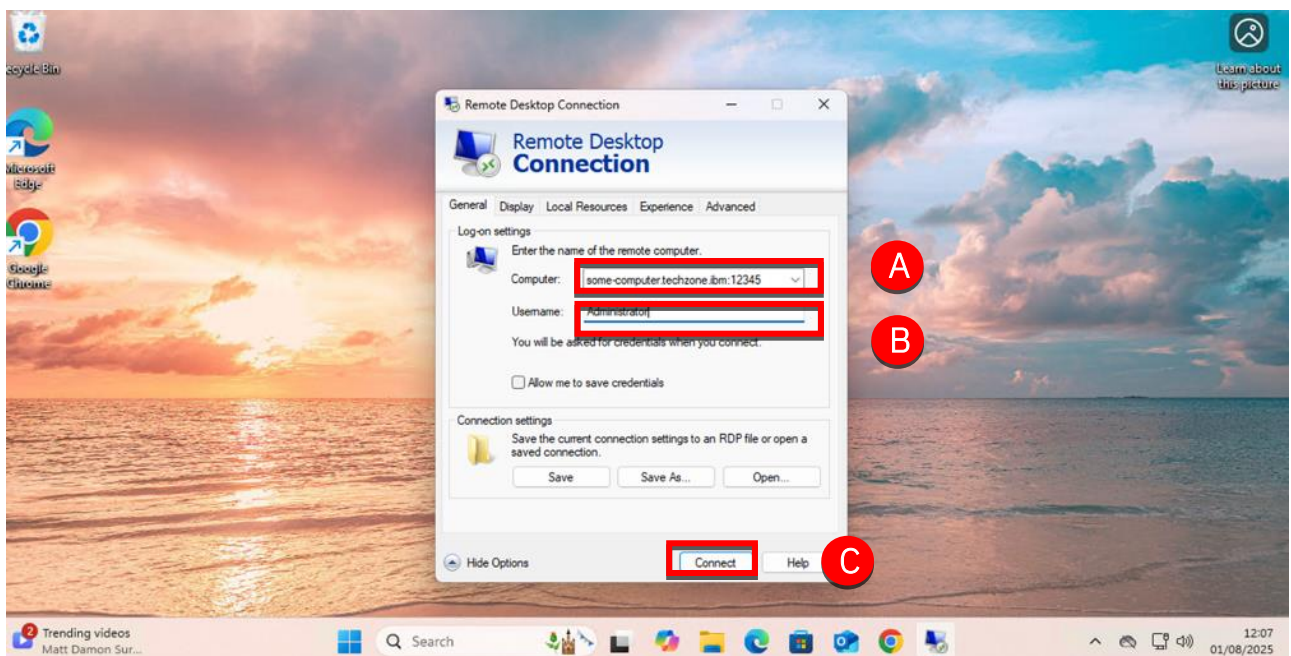
1. Start Remote Desktop Connection. Type “Remote Desktop” into the search bar (A), then click **Remote Desktop Connection** (B).



2. Click **Show Options** (A).

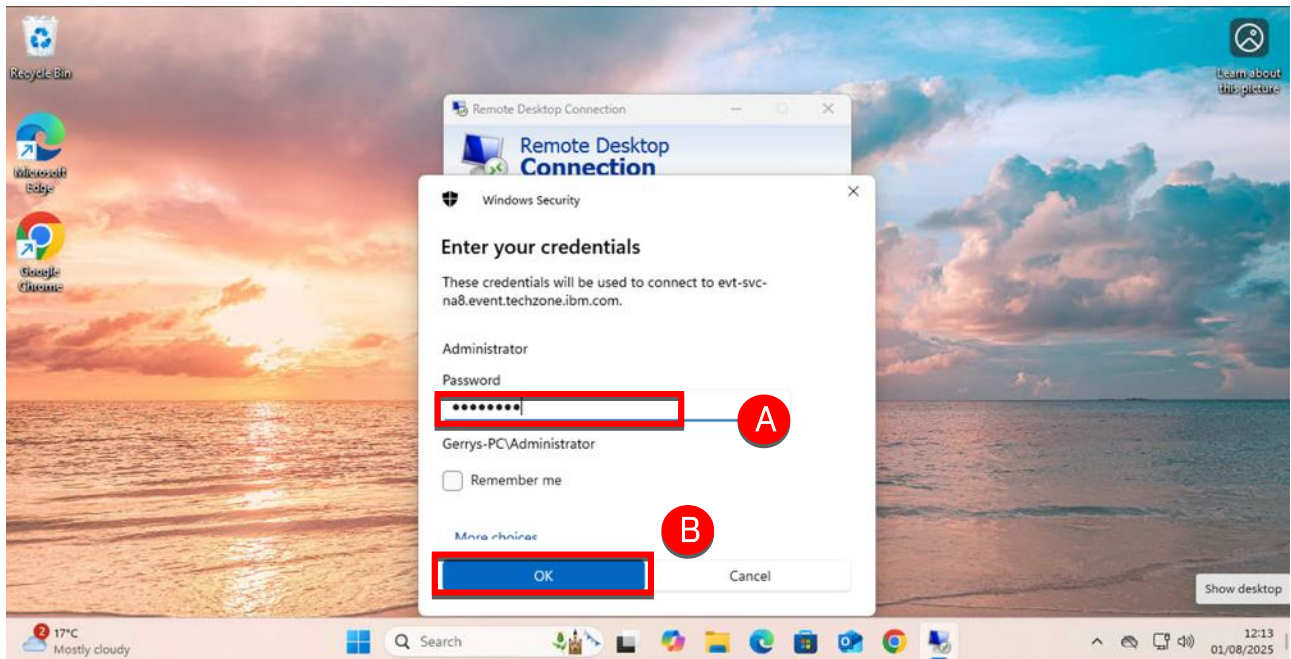


3. Enter the connection details for your reservation into the **Computer** field (Item 3) (A) and type "Administrator" into **Username** (B). Click **Connect** (C).

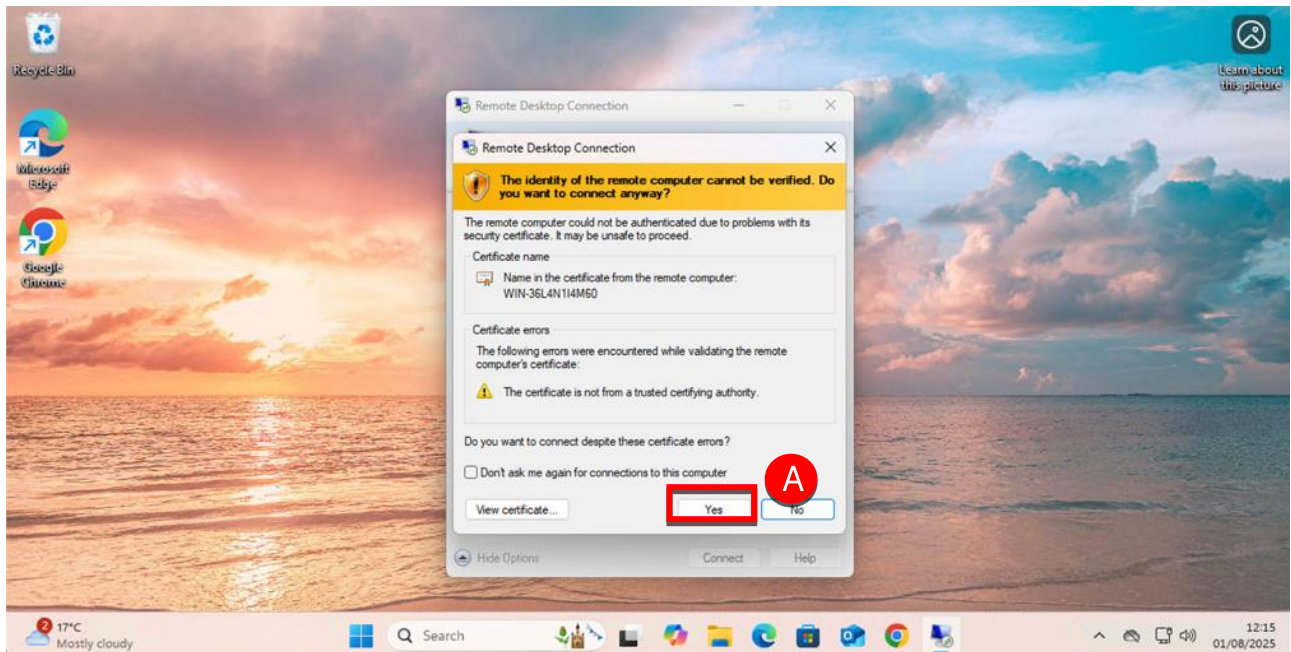


If you encounter a credentials error, use the username Administrator. If this also fails, try the username \\Administrator.

4. Type “IBMDem0s” in **Password** (A). Click **OK** (B).



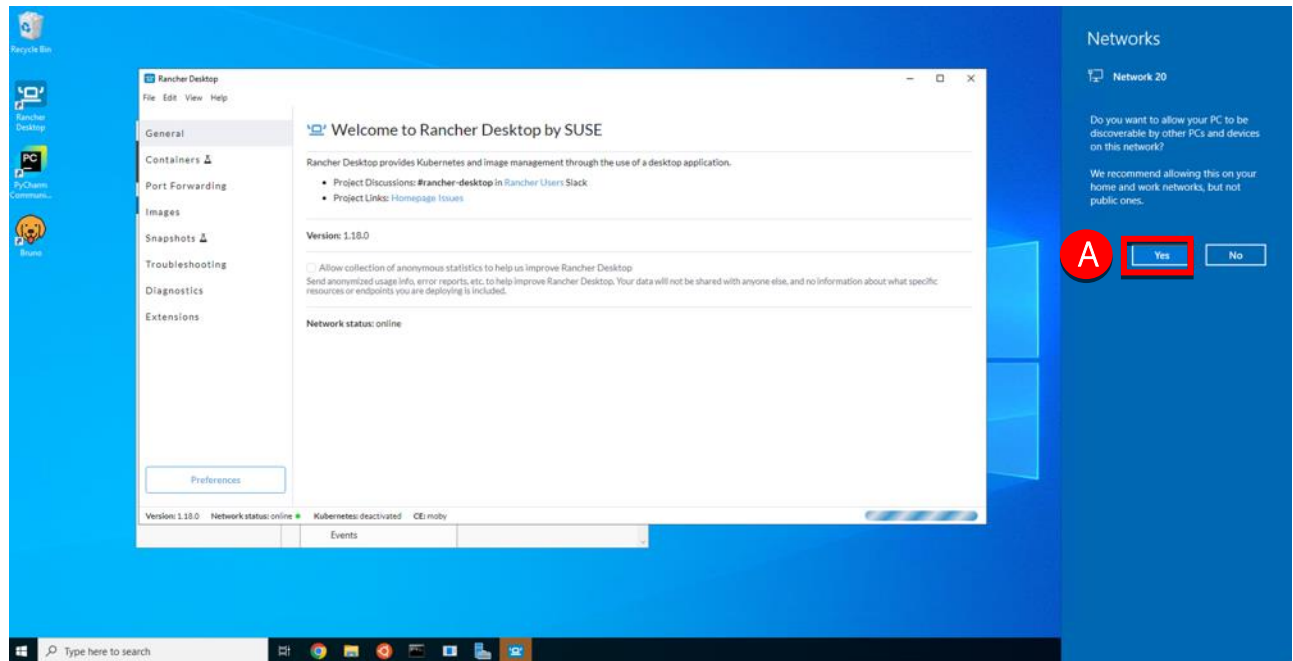
5. Click **Yes** (A) to connect, ignoring the certificate error.



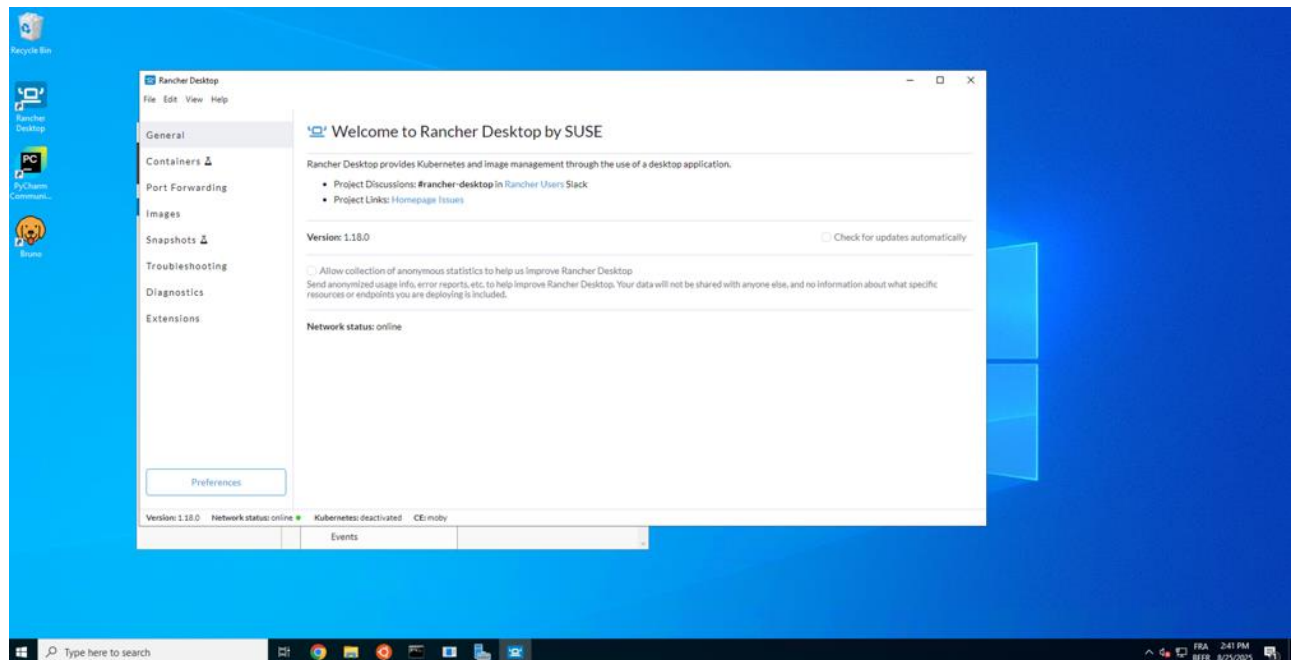
Application startup

It will take a few minutes for startup applications to be ready.

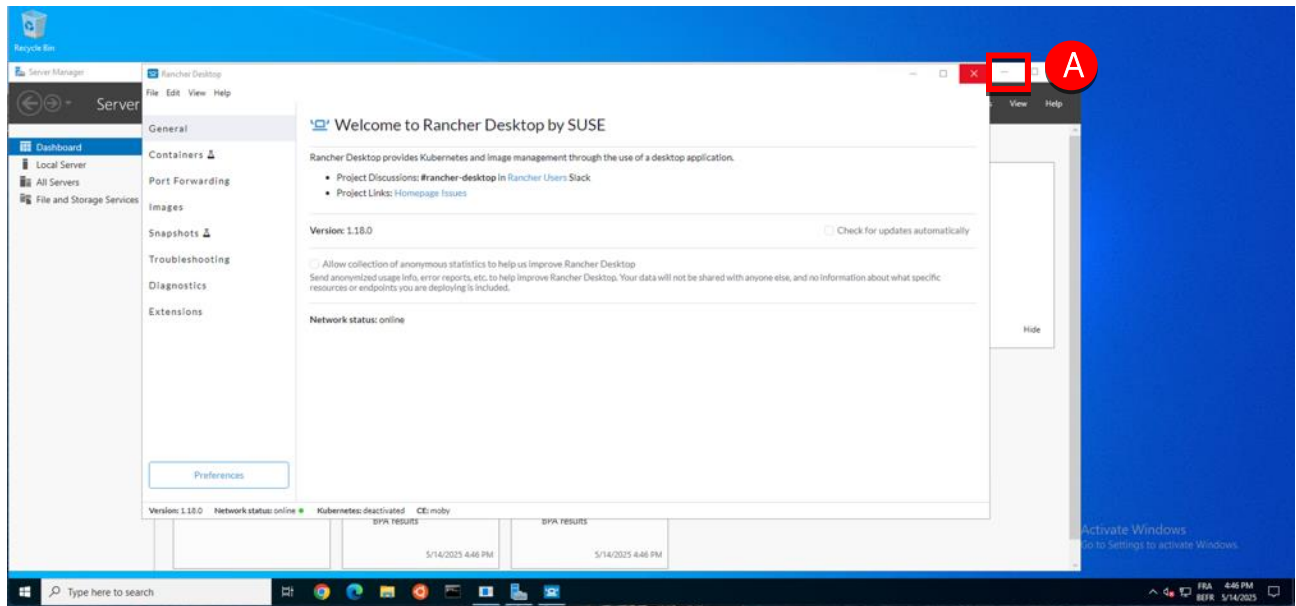
1. After a few minutes, your image should be started. In the **Networks** section, click Yes (A).



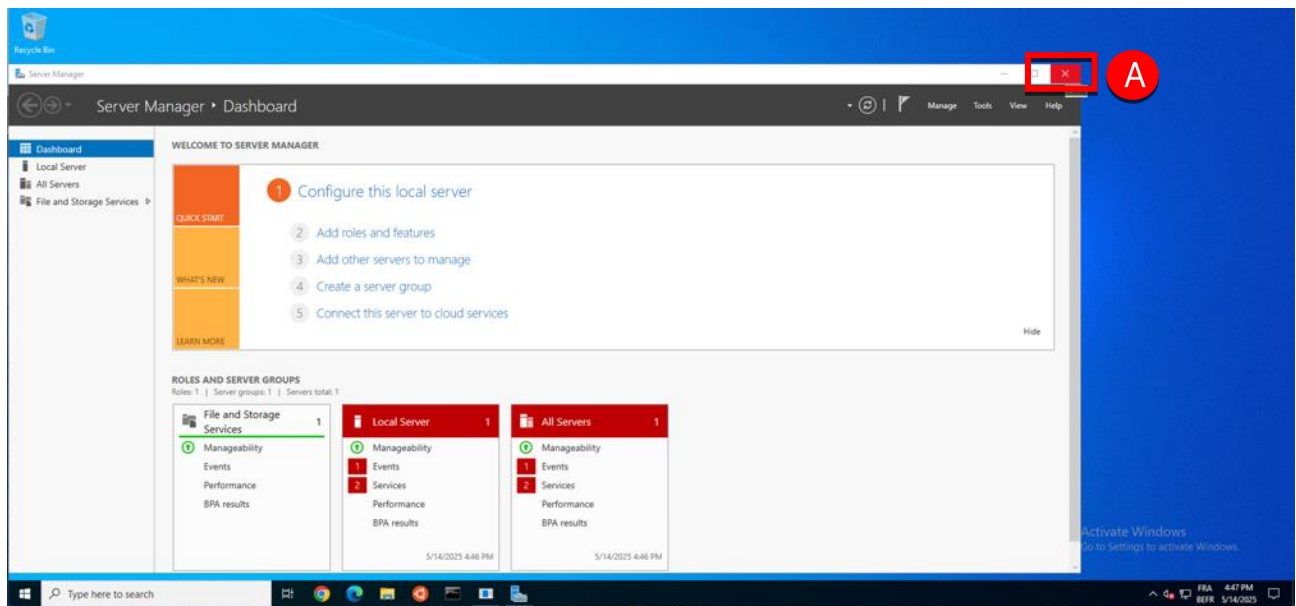
2. Wait for Rancher Desktop to be fully started (A) (no more activity listed).



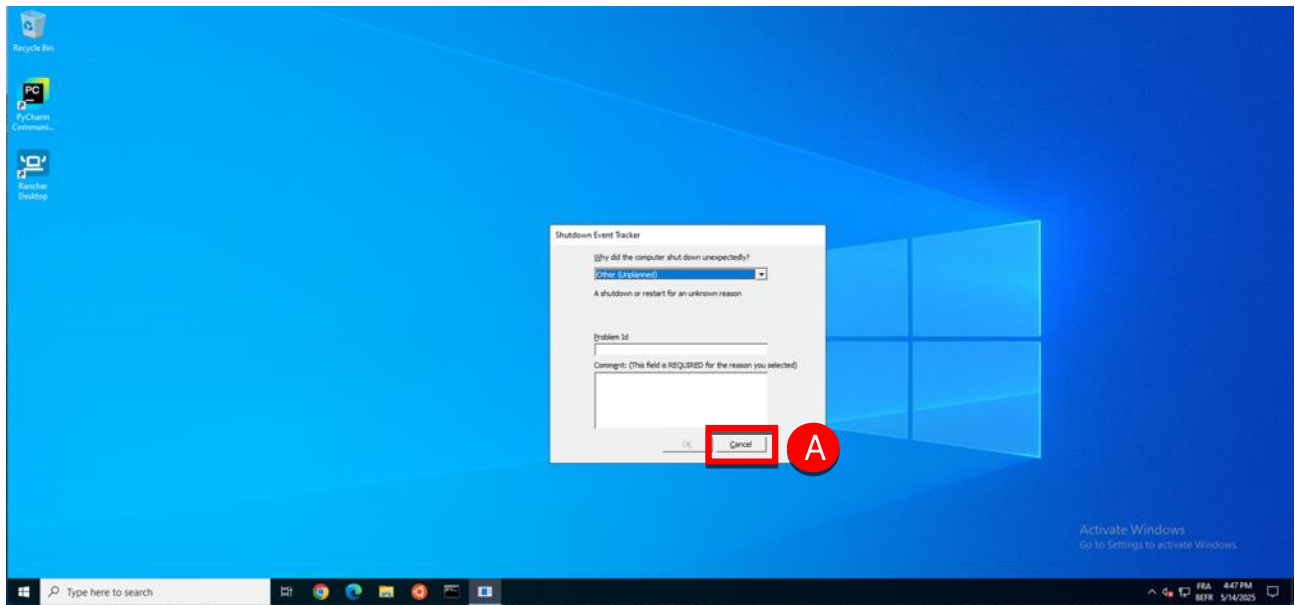
3. Click the icon to minimize the **Welcome Rancher Desktop** window (A). Note, even if closed the Rancher services will remain running.



4. Click X to close the **Server Manager** window (A)

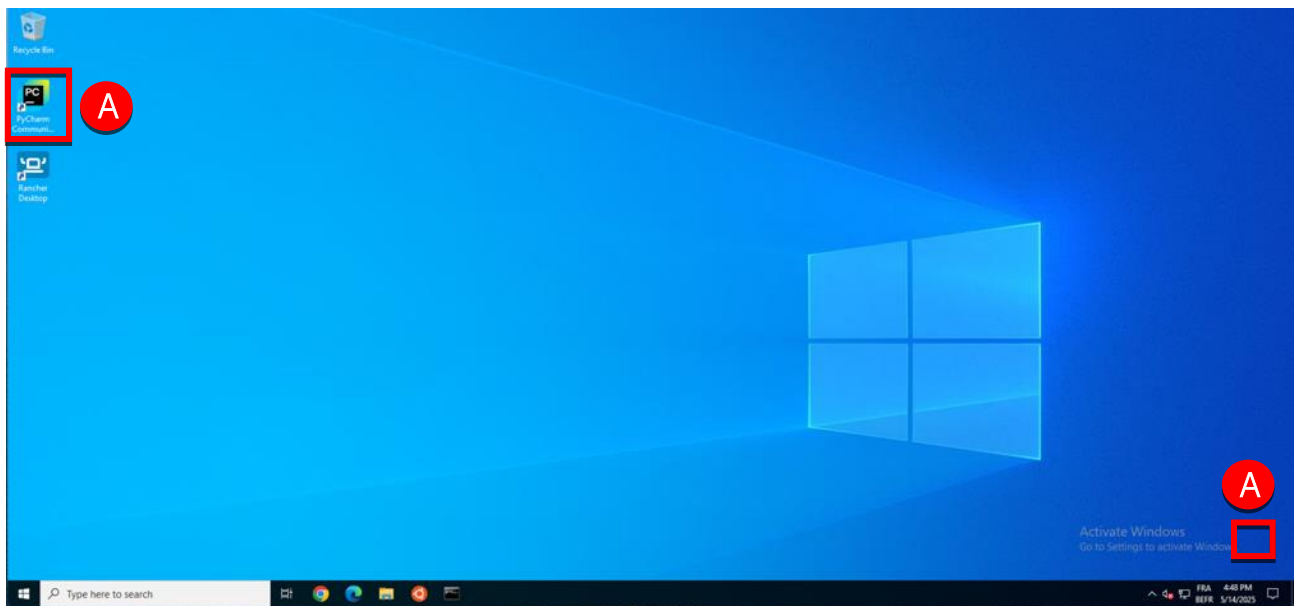


5. Click **Cancel** to close the Shutdown Event Tracker (A) (if required).

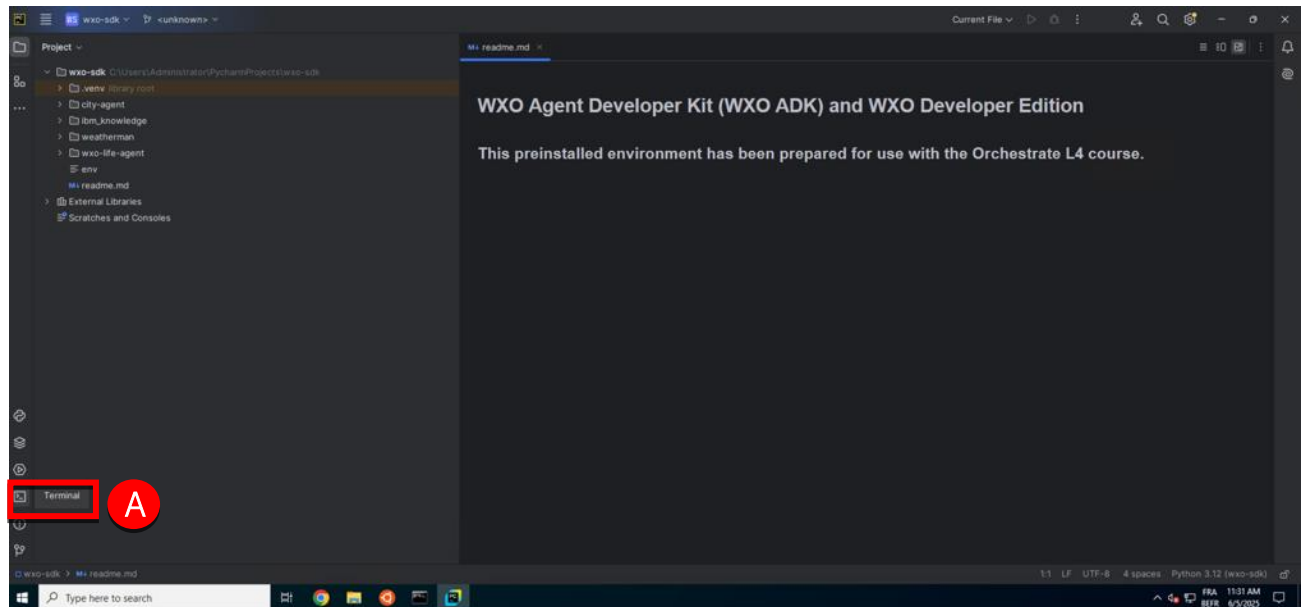


Configure your Orchestrate server

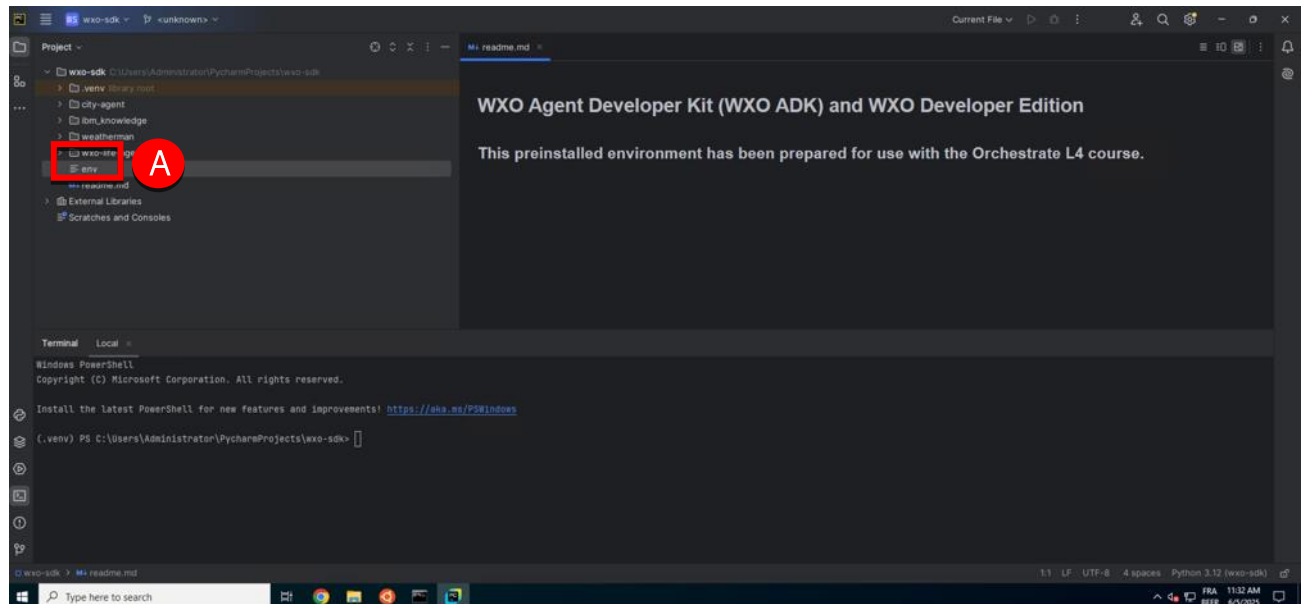
1. Double-click the **PyCharm** icon to open the **PyCharm Community edition** application.



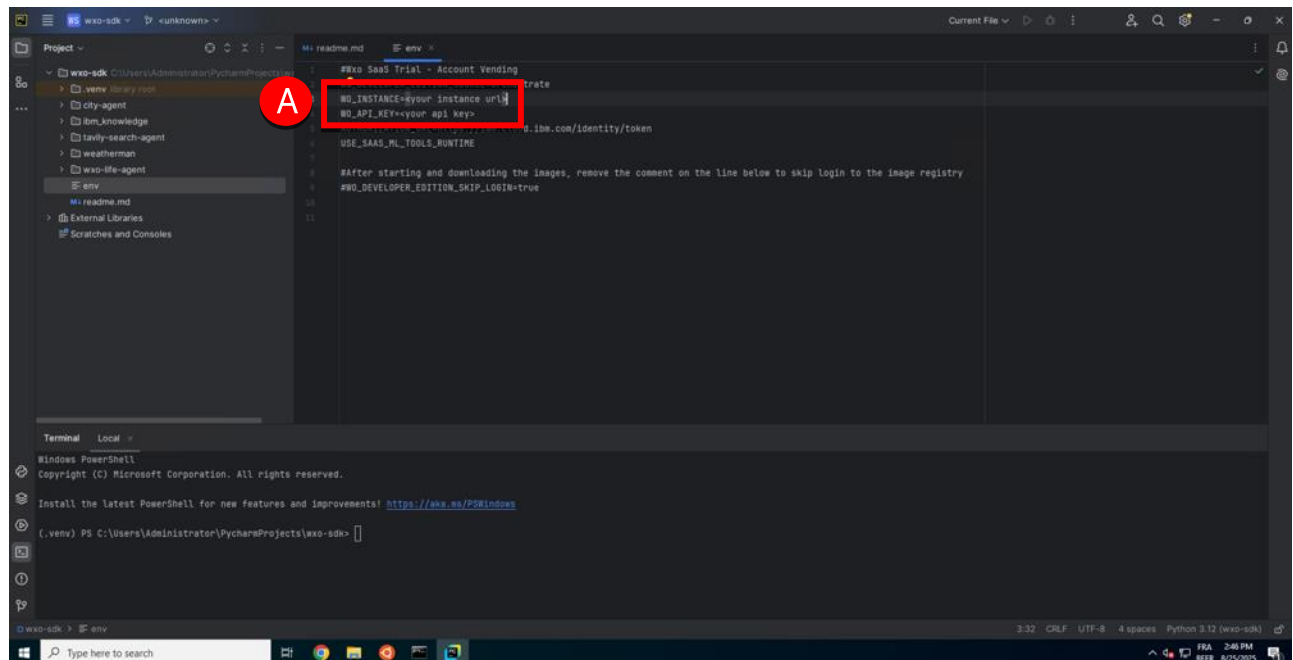
- Click the **Terminal** icon to open the terminal panel (A) (if not already opened).



- Double click the **env** file (A) in the **wso_sdk** folder to open the environment configuration.



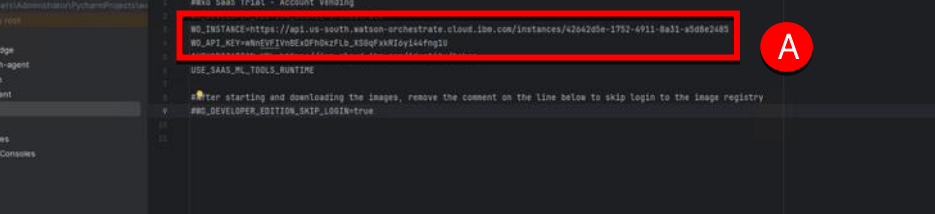
The env file must contain the configuration parameters required to start the watsonx Orchestrate server. The parameters values (A) are the ones that have been collected and copied into your notes in the previous steps.



- **WO_INSTANCE** is the service URL you have collected from your SaaS environment ([get your instance URL](#)) and stored in item 7 in your note.
 - **WO_API_KEY** is the API key you have created in your IBM Cloud environment ([create your API key](#)) and stored in item 8 in your note.
4. Update the following text with your own keys (A) and copy the entire text to your clipboard.

```
WO_INSTANCE=<your SaaS service url>  
WO_API_KEY=<your IBM Cloud API key>
```

A

- 
- The screenshot shows a Windows PowerShell terminal window with the following content:
- ```

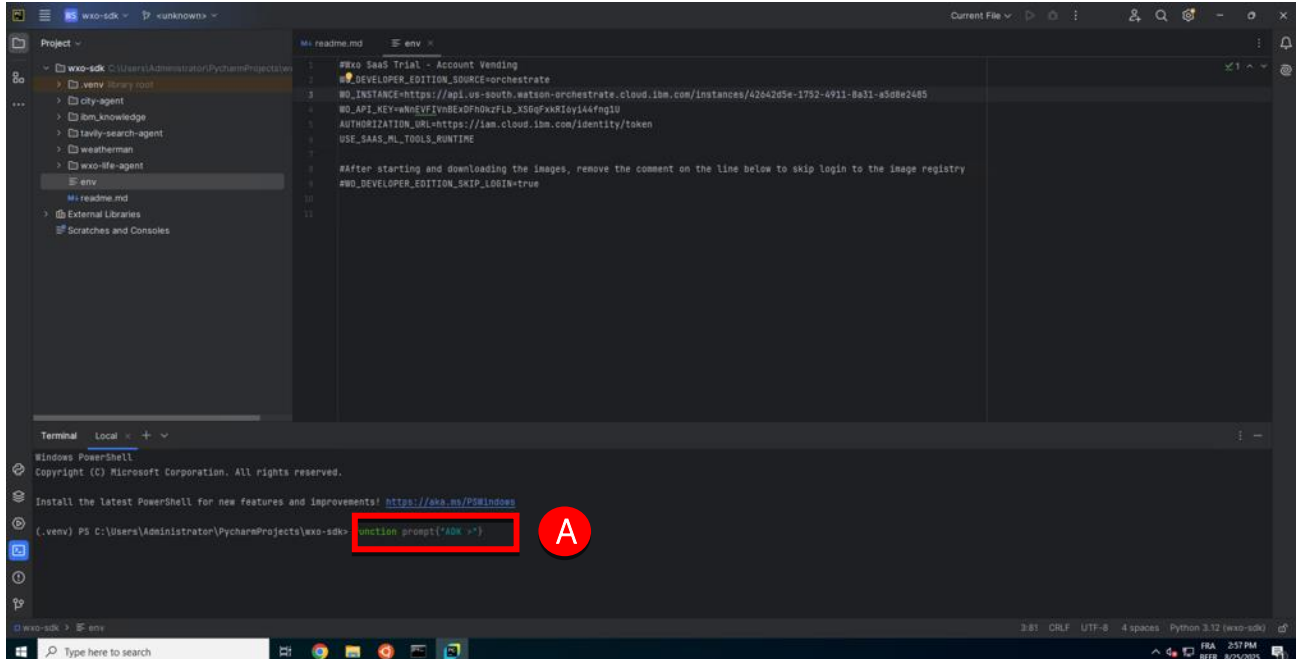
PS C:\Users\Administrator\PycharmProjects\wso-sdk> pip install wso-sdk
Collecting wso-sdk
 Downloading wso_sdk-0.0.1-py3-none-any.whl (1.1 MB)
 ...
Installing collected packages: wso-sdk
Successfully installed wso-sdk-0.0.1

```
- A red circle with a white letter 'A' is overlaid on the terminal window.

Watsonx Orchestrate – Level 4 lab prep guide

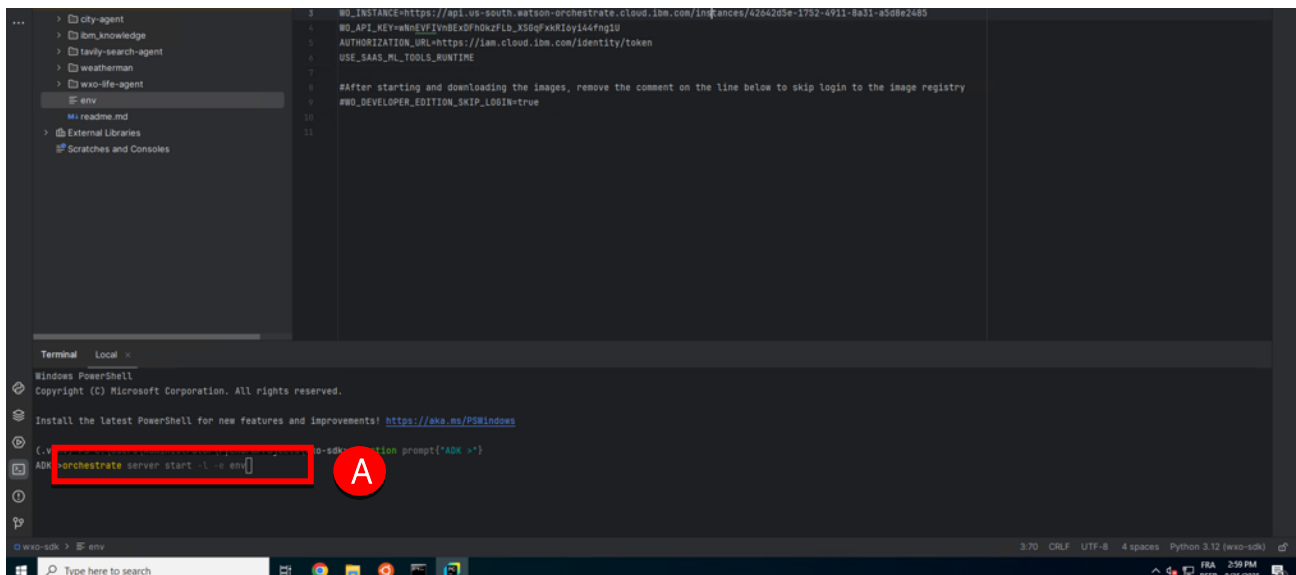
6. Enter the following command into the terminal to improve the command visibility, then press Enter:

```
function prompt{"ADK >"}
```



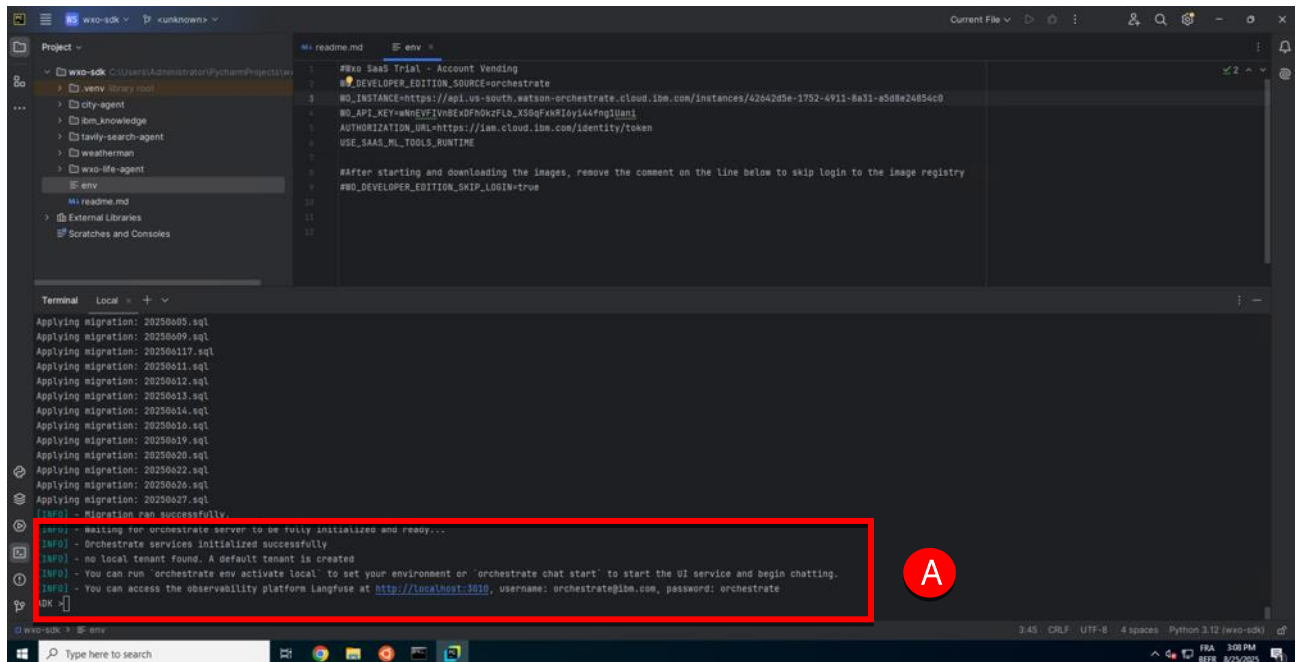
7. Enter the following command into the terminal to start the server with your env configuration, then press Enter.

```
orchestrate server start -l -e env
```



**Note:** Some services may fail to start on the first attempt, and you may encounter issues connecting to the docker repository. Retry the start command in this situation, depending on the volume of requests, it can take several attempts to connect.

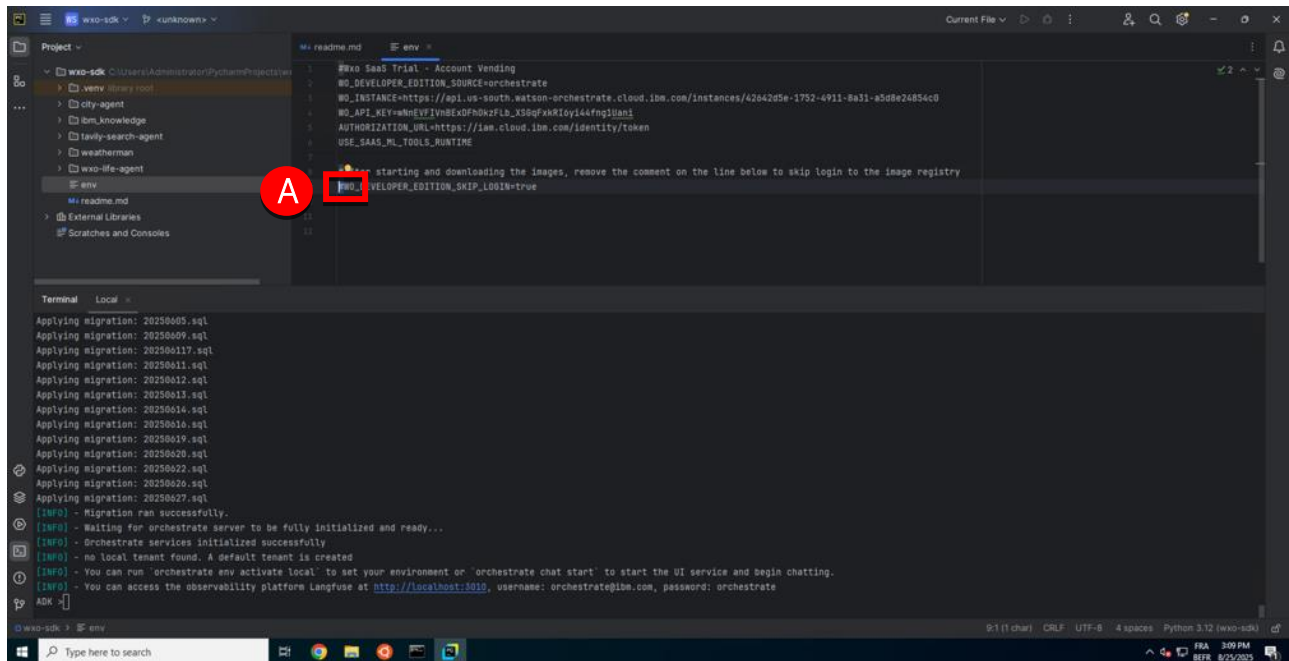
When startup is complete the terminal output will resemble the example below, restart can take 2-3 minutes to complete. The local server will be ready when you will see this **INFO** in the terminal screen (A):



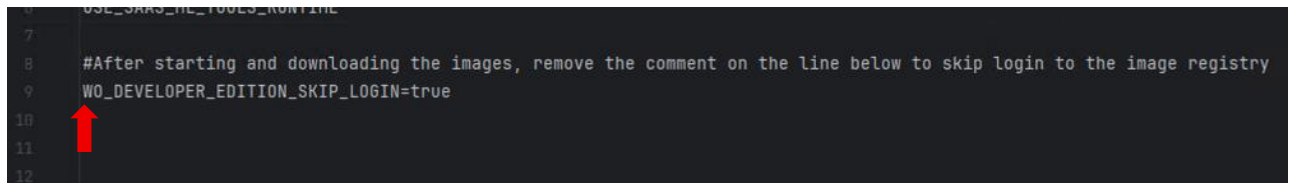
```
Project - wso-sdk - C:\Users\Administrator\PycharmProjects\wso-sdk
...
> .venv\library\tools
> city-agent
> iam-knowledge
> tavily-search-agent
> weatherman
> wso-lfe-agent
> .env
> External Libraries
> Scratches and Consoles

Terminal Local +
Applying migration: 20250605.sql
Applying migration: 20250609.sql
Applying migration: 202506117.sql
Applying migration: 20250611.sql
Applying migration: 20250612.sql
Applying migration: 20250613.sql
Applying migration: 20250614.sql
Applying migration: 20250616.sql
Applying migration: 20250619.sql
Applying migration: 20250620.sql
Applying migration: 20250622.sql
Applying migration: 20250626.sql
Applying migration: 20250627.sql
[INFO] - Migration ran successfully.
[INFO] - Waiting for orchestrate server to be fully initialized and ready...
[INFO] - Orchestrate services initialized successfully
[INFO] - no local tenant found. A default tenant is created
[INFO] - You can run 'orchestrate env activate local' to set your environment or 'orchestrate chat start' to start the UI service and begin chatting.
[INFO] - You can access the observability platform Langfuse at http://localhost:3030, username: orchestrate@ibm.com, password: orchestrate
OK
```

- Remove the comment (# character) from the env file as specified in the file instruction (A).

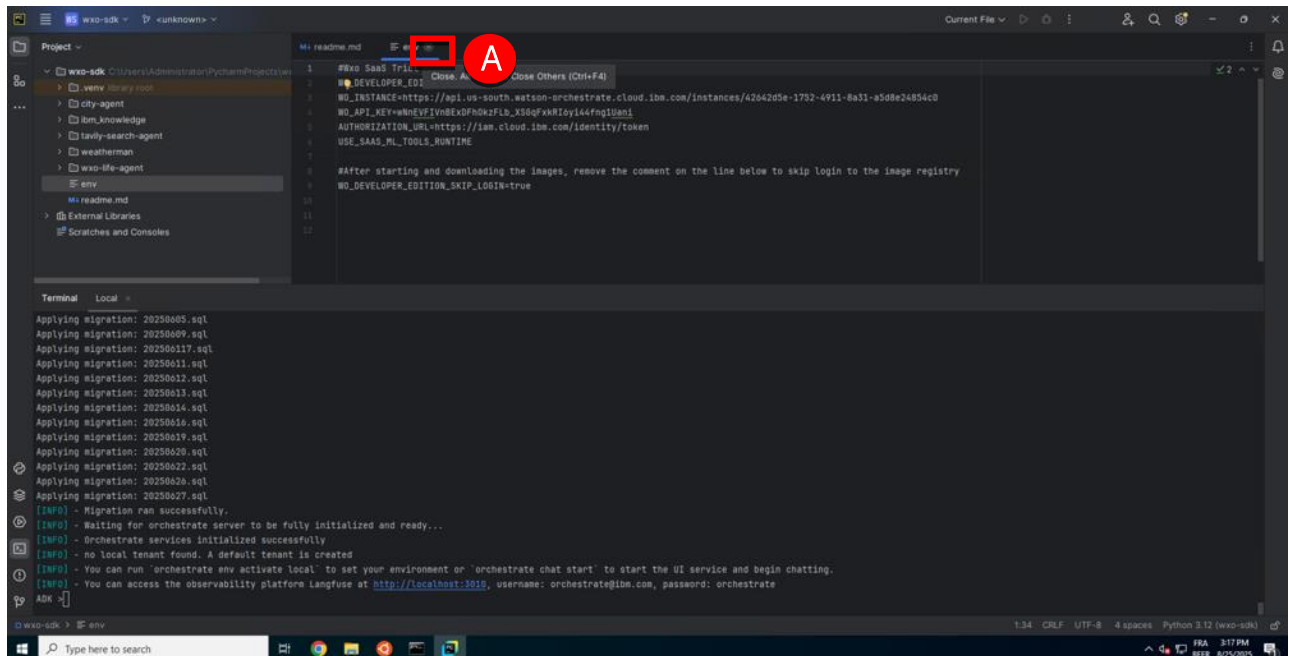


The last line in the env file should resemble the example below:



Now that the docker images have been downloaded, this setting skips the login into the image repository.

9. Click X to close the env file (A).



You are now ready to work with watsonx Orchestrator ADK.

### Additional orchestrate ADK commands:

If you need to stop, or reset your server, or stop the chat server, the following commands can be used. Note that these steps are for information only, and do not need to be performed here.

If you need to stop your server, enter the following command into the terminal, then press Enter. (Don't do this if you have just setup your environment as you will need the server running).

```
orchestrate server stop
```

To reset your server and stop it, enter the following command into the terminal, then press Enter.

```
orchestrate server reset
```

To stop your chat service, enter the following command into the terminal, then press Enter.

```
orchestrate chat stop
```

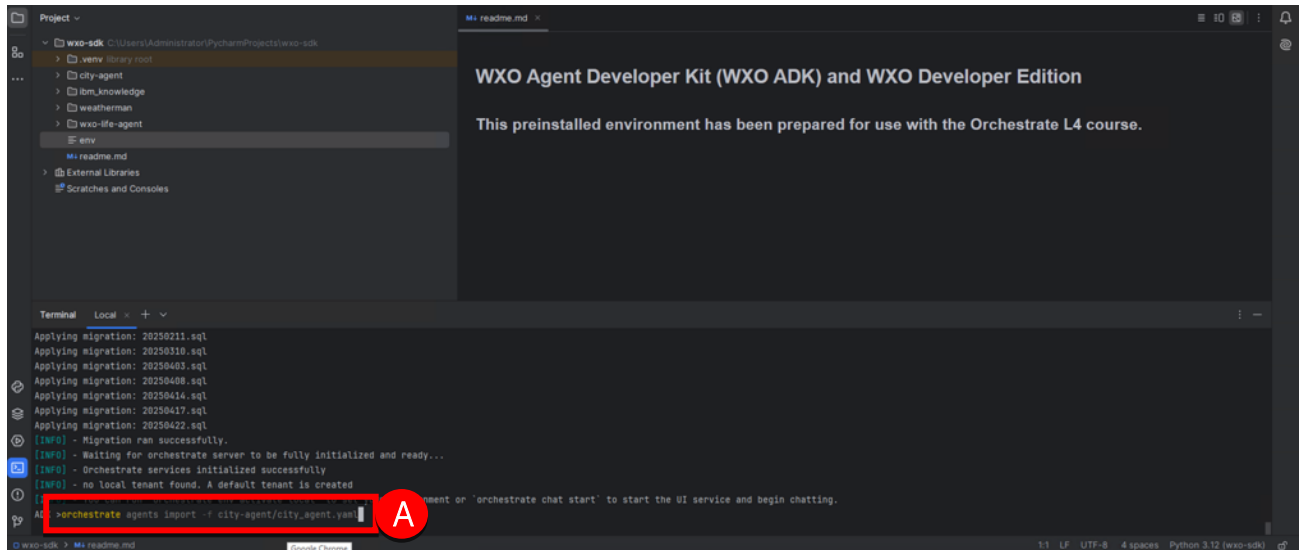
### Validate your environment

In this section you will deploy a simple agent and test it using the watsonx Orchestrator chat interface. This step is just to validate that your environment is correctly configured to perform the other labs. The

configuration of the agent will not be explained in this section, it's only purpose is to validate that your environment has been configured correctly. These instructions assume your server is running.

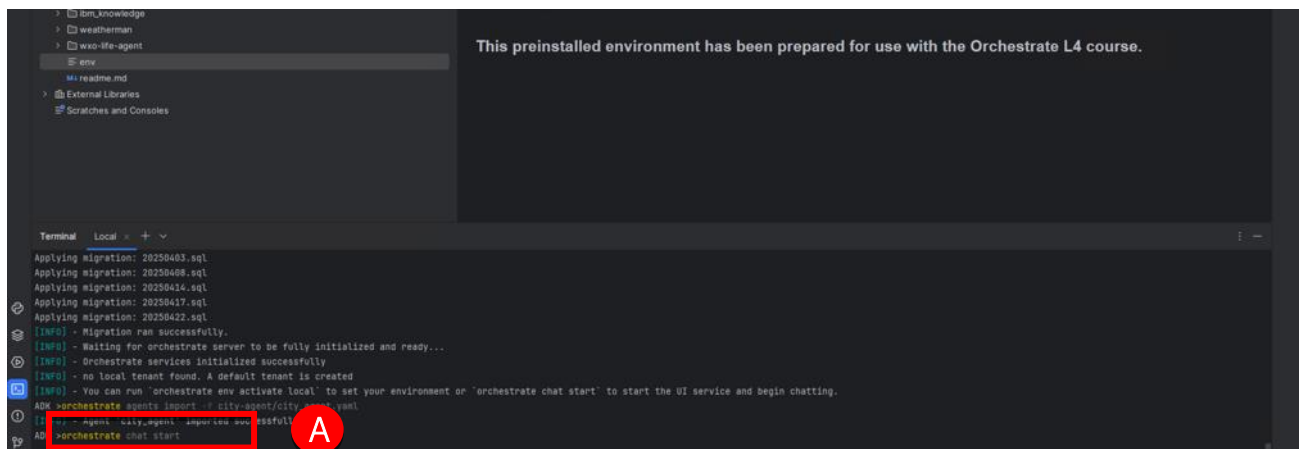
1. Enter the following command into the terminal (A) to create an agent.

```
orchestrate agents import -f city-agent/city_agent.yaml
```



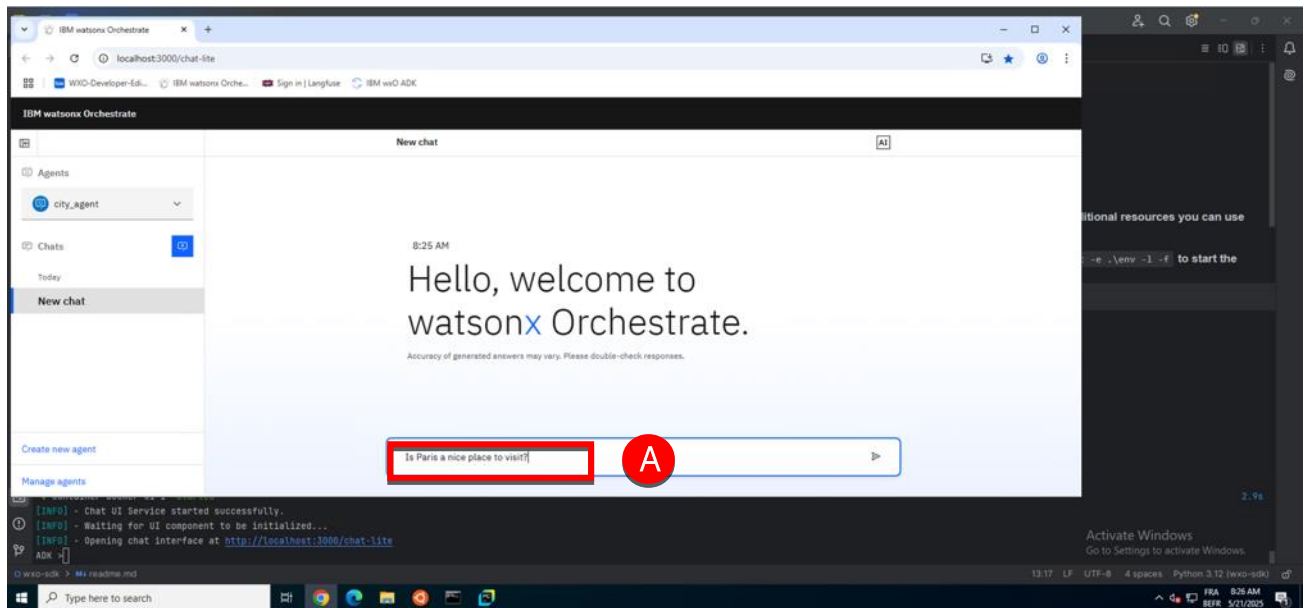
2. Enter the following command into the terminal (A) to start the chat server.

```
orchestrate chat start
```

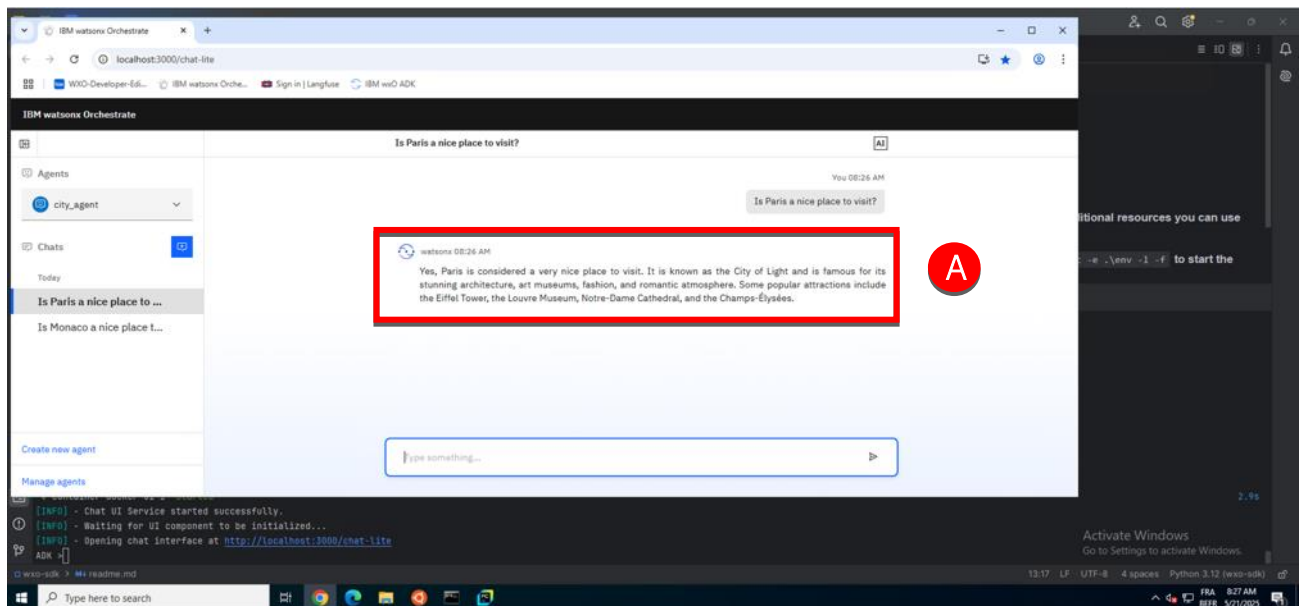




- When the browser opens enter "Is Paris a nice place to visit?" into the chat window (A), then press Enter.



- Wait for the answer (A).



If the system responded with an answer, the agent is working correctly, and the system is configured correctly to work with the foundation models deployed in watsonx.ai.

## Getting help

If you require assistance in interpreting any of the steps in this lab, please post your questions to the **#ba-techlcd-support** Slack channel (IBMer only).

Business Partners can request help at the [Partner Plus Support](#) website.

For troubleshooting tips, see the **TechZone Set Up Troubleshooting Guide** (<https://ibm.seismic.com/Link/Content/DCGT3pQ7hHM828WDQ86R7Tf6gpPV>) for help with common issues and solutions when using IBM Technology Zone. Additionally, go to the IBM Technology Zone Help page (<https://techzone.ibm.com/help>).

If you have an issue with the site, you can open a support case ([https://ibmsf.force.com/ibminternalproducts/s/createrecord/NewCase?language=en\\_US](https://ibmsf.force.com/ibminternalproducts/s/createrecord/NewCase?language=en_US)). Alternatively, you can send an email to [techzone.help@ibm.com](mailto:techzone.help@ibm.com). This contact information is also provided in the emails that you receive from IBM Technology Zone.

Additionally, see the watsonx.ai documentation (<https://dataplatfom.cloud.ibm.com/docs/content/wsi/getting-started/welcome-main.html?context=wx&audience=wdp>).

Or the watsonx Orchestrate DE documentation ([https://developer.watson-orchestrate.ibm.com/getting\\_started/wxOde\\_setup](https://developer.watson-orchestrate.ibm.com/getting_started/wxOde_setup))

Or the watsonx Orchestrate documentation (<https://www.ibm.com/docs/en/watsonx/watson-orchestrate/base?topic=getting-started-watsonx-orchestrate>)



